## §3. PRODUCTS OF PRESCHEMES

**3.1. Sums of preschemes** Let $(X_\alpha)$ be any family of preschemes; let $X$ be a topological space which is the *sum* of the underlying spaces $X_\alpha$; $X$ is then the union of pairwise disjoint open subspaces $X'_\alpha$, and for each $\alpha$ there is a homeomorphism $\phi_\alpha$ from $X_\alpha$ to $X'_\alpha$. If we equip each of the $X'_\alpha$ with the sheaf $(\phi_\alpha)_*(\mathscr{O}_{X_\alpha})$, it is clear that $X$ becomes a prescheme, which we call the *sum* of the family of preschemes $(X_\alpha)$ and which we denote $\bigsqcup_\alpha X_\alpha$. If $Y$ is a prescheme, then the map $f \mapsto (f \circ \phi_\alpha)$ is a *bijection* from the set $\operatorname{Hom}(X, Y)$ to the product set $\prod_\alpha \operatorname{Hom}(X_\alpha, Y)$. In particular, if the $X_\alpha$ are $S$-preschemes, with structure morphisms $\psi_\alpha$, then $X$ is an $S$-prescheme by the unique morphism $\psi : X \to S$ such that $\psi \circ \phi_\alpha = \psi_\alpha$ for each $\alpha$. The sum of two preschemes $X$ and $Y$ is denoted by $X \sqcup Y$. It is immediate that, if $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$, then $X \sqcup Y$ is canonically identified with $\operatorname{Spec}(A \times B)$.

**3.2. Products of preschemes**

**Definition (3.2.1). —** Given $S$-preschemes $X$ and $Y$, we say that a triple $(Z, p_1, p_2)$, consisting of an $S$-prescheme $Z$, and $S$-morphisms $p_1 : Z \to X$ and $p_2 : Z \to Y$, is a product of the $S$-preschemes **I** | 105
$X$ and $Y$, if, for each $S$-prescheme $T$, the map $f \mapsto (p_1 \circ f, p_2 \circ f)$ is a bijection from the set of $S$-morphisms from $T$ to $Z$, to the set of pairs consisting of an $S$-morphism $T \to X$ and an $S$-morphism $T \to Y$ (in other words, a bijection

$$\operatorname{Hom}_S(T, Z) \simeq \operatorname{Hom}_S(T, X) \times \operatorname{Hom}_S(T, Y)).$$

This is the general notion of a *product* of two objects in a category, applied to the category of $S$-preschemes (T, I, 1.1); in particular, a product of two $S$-preschemes is *unique* up to a unique $S$-isomorphism. Because of this uniqueness, most of the time we will denote a product of two $S$-preschemes $X$ and $Y$ by $X \times_S Y$ (or simply $X \times Y$, when there is no chance of confusion), with the morphisms $p_1$ and $p_2$ (the *canonical projections* of $X \times_S Y$ to $X$ and to $Y$, respectively) being suppressed in the notation. If $g : T \to X$ and $h : T \to Y$ are $S$-morphisms, we denote by $(g, h)_S$, or simply $(g, h)$, the $S$-morphism $f : T \to X \times_S Y$ such that $p_1 \circ f = g$, $p_2 \circ f = h$. If $X'$ and $Y'$ are two $S$-preschemes, $p'_1$ and $p'_2$ the canonical projections of $X' \times_S Y'$ (assumed to exist), and $u : X' \to X$ and $v : Y' \to Y$ are $S$-morphisms, we write $u \times_S v$ (or simply $u \times v$) for the $S$-morphism $(u \circ p'_1, v \circ p'_2)_S$ from $X' \times_S Y'$ to $X \times_S Y$.

When $S$ is an affine scheme given by some ring $A$, we often replace $S$ by $A$ in the above notation.

**Proposition (3.2.2). —** *Let $X$, $Y$, and $S$ be affine schemes, given by rings $B$, $C$, and $A$ (respectively). Let $Z = \operatorname{Spec}(B \otimes_A C)$, and let $p_1$ and $p_2$ be the $S$-morphisms corresponding* (2.2.4) *to the canonical $A$-homomorphisms $u : b \mapsto b \otimes 1$ and $v : c \mapsto 1 \otimes c$ (respectively) from $B$ and $C$ to $B \otimes_A C$; then $(Z, p_1, p_2)$ is a product of $X$ and $Y$.*

*Proof.* According to (2.2.4), it suffices to check that, if, to each $A$-homomorphism $f : B \otimes_A C \to L$ (where $L$ is an $A$-algebra), we associate the pair $(f \circ u, f \circ v)$, then this defines a bijection $\operatorname{Hom}_A(B \otimes_A C, L) \simeq \operatorname{Hom}_A(B, L) \times \operatorname{Hom}_A(C, L)$,[1] which follows immediately from the definitions and the fact that $b \otimes c = (b \otimes 1)(1 \otimes c)$. $\square$

**Corollary (3.2.3). —** *Let $T$ be an affine scheme given by some ring $D$, and $\alpha = ({}^a\rho, \widetilde{\rho})$ (resp. $\beta = ({}^a\sigma, \widetilde{\sigma})$) an $S$-morphism $T \to X$ (resp. $T \to Y$), where $\rho$ (resp. $\sigma$) is an $A$-homomorphism from $B$ (resp. $C$) to $D$; then $(\alpha, \beta)_S = ({}^a\tau, \widetilde{\tau})$, where $\tau$ is the homomorphism $B \otimes_A C \to D$ such that $\tau(b \otimes c) = \rho(b)\sigma(c)$.*

**Proposition (3.2.4). —** *Let $f : S' \to S$ be a* monomorphism *of preschemes* (T, I, 1.1)*, and let $X$ and $Y$ be $S'$-preschemes, also considered as $S$-preschemes via $f$. Every product of the $S$-preschemes $X$ and $Y$ is then a product of the $S'$-preschemes $X$ and $Y$, and vice versa.*

*Proof.* Let $\phi : X \to S'$ and $\psi : Y \to S'$ be the structure morphisms. If $T$ is an $S$-prescheme, and $u : T \to X$ and $v : T \to Y$ are $S$-morphisms, then we have, by definition, that $f \circ \phi \circ u = f \circ \psi \circ v = \theta$, the structure morphism of $T$; the hypothesis on $f$ implies that $\phi \circ u = \psi \circ v = \theta'$, and so we can consider $T$ as an $S'$-prescheme with structure morphism $\theta'$, and $u$ and $v$ as $S'$-morphisms. The conclusion of the proposition follows immediately, taking (3.2.1) into account. $\square$

[1] The notation $\operatorname{Hom}_A$ denotes here the set of homomorphisms of *A-algebras*.

**Corollary (3.2.5). —** *Let $X$ and $Y$ be $S$-preschemes, with structure morphisms $\phi : X \to S$ and $\psi : Y \to S$, and let $S'$ be an open subset of $S$ such that $\phi(X) \subset S'$ and $\psi(Y) \subset S'$. Every product of the $S$-preschemes $X$ and $Y$ is then also a product of the $S'$-preschemes $X$ and $Y$, and conversely.*

It suffices to apply (3.2.4) to the canonical injection $S' \to S$. **I** | 106

**Theorem (3.2.6). —** *Given $S$-preschemes $X$ and $Y$, there exists a product $X \times_S Y$.*

The proof proceeds in several steps.

**Lemma (3.2.6.1). —** *Let $(Z, p, q)$ be a product of $X$ and $Y$, and $U$ and $V$ open subsets of $X$ and $Y$, respectively. If we let $W = p^{-1}(U) \cap q^{-1}(V)$, then the triple consisting of $W$ and the restrictions of $p$ and $q$ to $W$* (considered as the morphisms $W \to U$ and $W \to V$, respectively) *is a product of $U$ and $V$.*

Indeed, if $T$ is an $S$-prescheme, then we can identify the $S$-morphisms $T \to W$ with the $S$-morphisms $T \to Z$ mapping $T$ to $W$. Then, if $g : T \to U$ and $h : T \to V$ are any two $S$-morphisms, we can consider them as $S$-morphisms from $T$ to $X$ and to $Y$, respectively, and, by hypothesis, there is then a unique $S$-morphism $f : T \to Z$ such that $g = p \circ f$ and $h = q \circ f$. Since $p(f(T)) \subset U$ and $q(f(T)) \subset V$, we have

$$f(T) \subset p^{-1}(U) \cap q^{-1}(V) = W,$$

hence our claim.

**Lemma (3.2.6.2). —** *Let $Z$ be an $S$-prescheme, $p : Z \to X$ and $q : Z \to Y$ both $S$-morphisms, $(U_\alpha)$ an open cover of $X$, and $(V_\lambda)$ an open cover of $Y$. Suppose that, for each pair $(\alpha, \lambda)$, the $S$-prescheme $W_{\alpha\lambda} = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ and the restrictions of $p$ and $q$ to $W_{\alpha\lambda}$ form a product of $U_\alpha$ and $V_\lambda$. Then $(Z, p, q)$ is a product of $X$ and $Y$.*

We first show that, if $f_1$ and $f_2$ are $S$-morphisms $T \to Z$, then the equations $p \circ f_1 = p \circ f_2$ and $q \circ f_1 = q \circ f_2$ imply that $f_1 = f_2$. Indeed, $Z$ is the union of the $W_{\alpha\lambda}$, so the $f_1^{-1}(W_{\alpha\lambda})$ form an open cover of $T$, and similarly for $f_2^{-1}(W_{\alpha\lambda})$. In addition, we have

$$f_1^{-1}(W_{\alpha\lambda}) = f_1^{-1}(p^{-1}(U_\alpha)) \cap f_1^{-1}(q^{-1}(V_\lambda)) = f_2^{-1}(p^{-1}(U_\alpha)) \cap f_2^{-1}(q^{-1}(V_\lambda)) = f_2^{-1}(W_{\alpha\lambda})$$

by hypothesis, and it thus reduces to noting that the restrictions of $f_1$ and $f_2$ to $f_1^{-1}(W_{\alpha\lambda}) = f_2^{-1}(W_{\alpha\lambda})$ are identical for each pair of indices. But since these restrictions can be considered as $S$-morphisms from $f_1^{-1}(W_{\alpha\lambda})$ to $W_{\alpha\lambda}$, our claim follows from the hypotheses and Definition (3.2.1).

Suppose now that we are given $S$-morphisms $g : T \to X$ and $h : T \to Y$. Let $T_{\alpha\lambda} = g^{-1}(U_\alpha) \cap h^{-1}(V_\lambda)$; then the $T_{\alpha\lambda}$ form an open cover of $T$. By hypothesis, there exists an $S$-morphism $f_{\alpha\lambda}$ such that $p \circ f_{\alpha\lambda}$ and $q \circ f_{\alpha\lambda}$ are, respectively, the restrictions of $g$ and $h$ to $T_{\alpha\lambda}$. We will now show that the restrictions of $f_{\alpha\lambda}$ and $f_{\beta\mu}$ to $T_{\alpha\lambda} \cap T_{\beta\mu}$ coincide, which will finish the proof of Lemma (3.2.6.2). The images of $T_{\alpha\lambda} \cap T_{\beta\mu}$ under $f_{\alpha\lambda}$ and $f_{\beta\mu}$ are contained in $W_{\alpha\lambda} \cap W_{\beta\mu}$ by definition. Since

$$W_{\alpha\lambda} \cap W_{\beta\mu} = p^{-1}(U_\alpha \cap U_\beta) \cap q^{-1}(V_\lambda \cap V_\mu),$$

it follows from Lemma (3.2.6.1) that $W_{\alpha\lambda} \cap W_{\beta\mu}$ and the restrictions to this prescheme of $p$ and $q$ form a *product* of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$. Since $p \circ f_{\alpha\lambda}$ and $p \circ f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$ and similarly for $q \circ f_{\alpha\lambda}$ and $q \circ f_{\beta\mu}$, we see that $f_{\alpha\lambda}$ and $f_{\beta\mu}$ coincide on $T_{\alpha\lambda} \cap T_{\beta\mu}$. □

**Lemma (3.2.6.3). —** *Let $(U_\alpha)$ be an open cover of $X$, $(V_\lambda)$ an open cover of $Y$, and suppose that, for each pair $(\alpha, \lambda)$, there exists a product of $U_\alpha$ and $V_\lambda$; then there exists a product of $X$ and $Y$.* **I** | 107

Applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$, we see that there exists a product of $S$-preschemes induced, respectively, by $X$ and $Y$ on these open sets; in addition, the uniqueness of the product shows that, if we set $i = (\alpha, \lambda)$ and $j = (\beta, \mu)$, then there is a canonical isomorphism $h_{ij}$ (resp. $h_{ji}$) from this product to an $S$-prescheme $W_{ij}$ (resp. $W_{ji}$) induced by $U_\alpha \times_S V_\lambda$ (resp. $U_\beta \times_S V_\mu$) on an open set; then $f_{ij} = h_{ij} \circ h_{ji}^{-1}$ is an isomorphism from $W_{ji}$ to $W_{ij}$. In addition, for a third pair $k = (\gamma, \nu)$, we have $f_{ik} = f_{ij} \circ f_{jk}$ on $W_{ki} \cap W_{kj}$, by applying Lemma (3.2.6.1) to the open sets $U_\alpha \cap U_\beta \cap U_\gamma$ and $V_\lambda \cap V_\mu \cap V_\nu$ in $U_\beta$ and $V_\mu$, respectively. It follows that we have a prescheme $Z$, an open cover $(Z_i)$ of the underlying space of $Z$, and, for each $i$, an isomorphism $g_i$ from the induced prescheme $Z_i$ to the prescheme $U_\alpha \times_S V_\lambda$, so that, for each pair $(i, j)$, we have

$f_{ij} = g_i \circ g_j^{-1}$ (2.3.1); in addition, we have $g_i(Z_i \cap Z_j) = W_{ij}$. Let $p_i$ and $q_i$ be the projections, and $\theta_i$ the structure morphism, of the $S$-prescheme $U_\alpha \times_S V_\lambda$. We immediately see that $p_i \circ g_i = p_j \circ g_j$ on $Z_i \cap Z_j$, and similarly for the two other morphisms. We can thus define the morphisms of preschemes $p : Z \to X$ (resp. $q : Z \to Y$, $\theta : Z \to S$) by the condition that $p$ (resp. $q$, $\theta$) coincide with $p_i \circ g_i$ (resp. $q_i \circ g_i$, $\theta_i \circ g_i$) on each of the $Z_i$; $Z$, equipped with $\theta$, is then an $S$-prescheme. We now show that $Z'_i = p^{-1}(U_\alpha) \cap q^{-1}(V_\lambda)$ is equal to $Z_i$. For each index $j = (\beta, \mu)$, we have $Z_j \cap Z'_i = g_j^{-1}(p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda))$. We have

$$p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = p_j^{-1}(U_\alpha \cap U_\beta) \cap q_j^{-1}(V_\lambda \cap V_\mu);$$

By Lemma (3.2.6.1), the restrictions of $p_j$ and $q_j$ to $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda)$ define, on this $S$-prescheme, the structure of a product of $U_\alpha \cap U_\beta$ and $V_\lambda \cap V_\mu$; but the uniqueness of the product then implies that $p_j^{-1}(U_\alpha) \cap q_j^{-1}(V_\lambda) = W_{ji}$. As a result, we have $Z_j \cap Z'_i = Z_j \cap Z_i$ for each $j$, hence $Z'_i = Z_i$. We then deduce from Lemma (3.2.6.2) that $(Z, p, q)$ is a product of $X$ and $Y$.

**Lemma (3.2.6.4).** — *Let $\phi : X \to S$ and $\psi : Y \to S$ be the structure morphisms of $X$ and $Y$, $(S_i)$ an open cover of $S$, and let $X_i = \phi^{-1}(S_i)$, $Y_i = \psi^{-1}(S_i)$. If each of the products $X_i \times_S Y_i$ exists, then $X \times_S Y$ exists.*

By Lemma (3.2.6.3), it suffices to prove that the products $X_i \times_S Y_j$ exist for all $i$ and $j$. Set $X_{ij} = X_i \cap X_j = \phi^{-1}(S_i \cap S_j)$, $Y_{ij} = Y_i \cap Y_j = \psi^{-1}(S_i \cap S_j)$; by Lemma (3.2.6.1), the product $Z_{ij} = X_{ij} \times_S Y_{ij}$ exists. We now note that, if $T$ is an $S$-prescheme, and if $g : T \to X_i$ and $h : T \to Y_j$ are $S$-morphisms, then we necessarily have that $\phi(g(T)) = \psi(h(T)) \subset S_i \cap S_j$ by the definition of an $S$-morphism, and thus that $g(T) \subset X_{ij}$ and $h(T) \subset Y_{ij}$; it is then immediate that $Z_{ij}$ is the product of $X_i$ and $Y_j$.

**(3.2.6.5).** We can now complete the proof of Theorem (3.2.6). If $S$ is an *affine scheme*, then there are covers $(U_\alpha)$ and $(V_\lambda)$ of $X$ and $Y$ (respectively) consisting of affine open subsets; since $U_\alpha \times_S V_\lambda$ exists by (3.2.2), Lemma (3.2.6.3) shows that $X \times_S Y$ exists. If $S$ is any prescheme, then there is a cover $(S_i)$ of $S$ consisting of affine open subsets. If $\phi : X \to S$ and $\psi : Y \to S$ are the structure morphisms, and if we set $X_i = \phi^{-1}(S_i)$ and $Y_i = \psi^{-1}(S_i)$, then the products $X_i \times_{S_i} Y_i$ exist, by the above; but then the products $X_i \times_S Y_i$ also exist (3.2.5), and Lemma (3.2.6.4) therefore shows that **I** | 108
$X \times_S Y$ exists. ☐

**Corollary (3.2.7).** — *Let $Z = X \times_S Y$ be the product of two $S$-preschemes, $p$ and $q$ the projections from $Z$ to $X$ and to $Y$ (respectively), and $\phi$ (resp. $\psi$) the structure morphism of $X$ (resp. $Y$). Let $S'$ be an open subset of $S$, and $U$ (resp. $V$) an open subset of $X$ (resp. $Y$) contained in $\phi^{-1}(S')$ (resp. $\psi^{-1}(S')$). Then the product $U \times_{S'} V$ is canonically identified with the prescheme induced on $Z$ by $p^{-1}(U) \cap q^{-1}(V)$ (considered as an $S'$-prescheme). In addition, if $f : T \to X$ and $g : T \to Y$ are $S$-morphisms such that $f(T) \subset U$ and $g(T) \subset V$, then the $S'$-morphism $(f, g)_{S'}$ can be identified with $(f, g)_S$ regarded as a morphism into $p^{-1}(U) \cap q^{-1}(V)$.*

*Proof.* This follows from Corollary (3.2.5) and Lemma (3.2.6.1). ☐

**(3.2.8).** Let $(X_\alpha)$ and $(Y_\lambda)$ be families of $S$-preschemes, and $X$ (resp. $Y$) the sum of the family $(X_\alpha)$ (resp. $(Y_\lambda)$) (3.1). Then $X \times_S Y$ can be identified with the *sum* of the family $(X_\alpha \times_S Y_\lambda)$; this follows immediately from Lemma (3.2.6.3).

**(3.2.9).** [2] It follows from (1.8.1) that we can state (3.2.2) in the following manner: $Z = \mathrm{Spec}(B \otimes_A C)$ **II** | 221
is not only a product of $X = \mathrm{Spec}(B)$ and $Y = \mathrm{Spec}(C)$ in the category of *$S$-preschemes*, but also in the category of *locally ringed spaces over $S$* (with a definition of $S$-morphisms modelled on that of (2.5.2)). The proof of (3.2.6) also proves that, for any two $S$-preschemes $X$ and $Y$, the prescheme $X \times_S Y$ is not only the product of $X$ and $Y$ in the category of $S$-preschemes, but also in the category of locally ringed spaces over the prescheme $S$.

[2] *[Trans.] (3.2.9) is from the errata of EGA II, on page 221, whence the change in page numbering.*

### 3.3. Formal properties of the product; change of the base prescheme

**(3.3.1).** The reader will notice that all the properties stated in this section, except (3.3.13) and (3.3.15), are true without modification in *any category*, whenever the products involved in the statements *exist* (since it is clear that the notions of an $S$-object and of an $S$-morphism can be defined exactly as in (2.5) for any object $S$ of the category).

**(3.3.2).** First of all, $X \times_S Y$ is a *covariant bifunctor* in $X$ and $Y$ on the category of $S$-preschemes: it suffices in fact to note that the diagram

$$\begin{tikzcd} X \times Y \arrow[r, "f\times 1"] \arrow[d] & X' \times Y \arrow[r, "f'\times 1"] \arrow[d] & X'' \times Y \arrow[d] \\ X \arrow[r, "f"] & X' \arrow[r, "f'"] & X'' \end{tikzcd}$$

is commutative.

**Proposition (3.3.3).** — *For each $S$-prescheme $X$, the first (resp. second) projection from $X \times_S S$ (resp. $S \times_S X$) is a functorial isomorphism from $X \times_S S$ (resp. $S \times_S X$) to $X$, whose inverse isomorphism is $(1_X, \phi)_S$ (resp. $(\phi, 1_X)_S$), where we denote by $\phi$ the structure morphism $X \to S$; we can therefore write, up to a canonical isomorphism,*

$$X \times_S S = S \times_S X = X.$$

*Proof.* It suffices to prove that the triple $(X, 1_X, \phi)$ is a product of $X$ and $S$. If $T$ is an $S$-prescheme, then the only $S$-morphism from $T$ to $S$ is necessarily the structure morphism $\psi : T \to S$. If $f$ is an $S$-morphism from $T$ to $X$, we necessarily have $\psi = \phi \circ f$, hence our claim. □

**Corollary (3.3.4).** — *Let $X$ and $Y$ be $S$-preschemes, with structure morphisms $\phi : X \to S$ and $\psi : Y \to S$. If we canonically identify $X$ with $X \times_S S$, and $Y$ with $S \times_S Y$, then the projections $X \times_S Y \to X$ and $X \times_S Y \to Y$ are identified with $1_X \times \psi$ and $\phi \times 1_Y$ (respectively).*

The proof is immediate and is left to the reader.

**(3.3.5).** We can define, in a manner similar to (3.2), the product of a finite number $n$ of $S$-preschemes, **I** | 109
and the existence of these products follows from (3.2.6) by induction on $n$, and by noting that $(X_1 \times_S X_2 \times_S \cdots \times_S X_{n-1}) \times_S X_n$ satisfies the definition of a product. The uniqueness of the product implies, as in any category, its *commutativity* and *associativity* properties. If, for example, $p_1$, $p_2$, and $p_3$ denote the projections from $X_1 \times_S X_2 \times_S X_3$, and if we identify this prescheme with $(X_1 \times_S X_2) \times_S X_3$, then the projection to $X_1 \times_S X_2$ is identified with $(p_1, p_2)_S$.

**(3.3.6).** Let $S$ and $S'$ be preschemes, and $\phi : S' \to S$ a morphism, which lets us consider $S'$ as an $S$-prescheme. For each $S$-prescheme $X$, consider the product $X \times_S S'$, and let $p$ and $\pi'$ be the projections to $X$ and to $S'$ (respectively). Equipped with $\pi'$, this product is an $S'$-prescheme; when we consider it as such, we denote it by $X_{(S')}$ or $X_{(\phi)}$, and we say that this is the prescheme obtained by *base change* (or *a change of base*) from $S$ to $S'$ by means of the morphism $\phi$, or the *inverse image* of $X$ by $\phi$. We note that, if $\pi$ is the structure morphism of $X$, and $\theta$ the structure morphism of $X \times_S S'$, considered as an $S$-prescheme, then the diagram

$$\begin{tikzcd} X \arrow[d, "\pi"'] & X_{(S')} \arrow[l, "p"'] \arrow[d, "\pi'"] \arrow[dl, "\theta"'] \\ S & S' \arrow[l, "\phi"] \end{tikzcd}$$

is commutative.

**(3.3.7).** With the notation of (3.3.6), for each $S$-morphism $f : X \to Y$, we denote by $f_{(S')}$ the $S'$-morphism $f \times_S 1 : X_{(S')} \to Y_{(S')}$, and we say that $f_{(S')}$ is the *base change* (or *inverse image*) of $f$ by $\phi$. Therefore $X_{(S')}$ is a *covariant functor* in $X$, from the category of $S$-preschemes to that of $S'$-preschemes.

**(3.3.8).** The prescheme $X_{(S')}$ can be considered as a solution to a *universal mapping problem*: each $S'$-prescheme $T$ is also an $S$-prescheme via $\phi$; each $S$-morphism $g : T \to X$ is then uniquely written as $g = p \circ f$, where $f$ is an $S'$-morphism $T \to X_{(S')}$, as follows from the definition of the product applied to the $S$-morphisms $g$[3] and $\psi : T \to S'$ (the structure morphism of $T$).

**Proposition (3.3.9). —** ("Transitivity of base change"). *Let $S''$ be a prescheme, and $\phi' : S'' \to S'$ a morphism. For each $S$-prescheme $X$, there exists a canonical functorial isomorphism from the $S''$-prescheme $(X_{(\phi)})_{(\phi')}$ to the $S''$-prescheme $X_{(\phi\circ\phi')}$.*

*Proof.* Let $T$ be a $S''$-prescheme, $\psi$ its structure morphism, and $g$ an $S$-morphism from $T$ to $X$ ($T$ being considered as an $S$-prescheme with structure morphism $\phi \circ \phi' \circ \psi$). Since $T$ is also an $S'$-prescheme with structure morphism $\phi' \circ \psi$, we can write $g = p \circ g'$, where $g'$ is an $S'$-morphism $T \to X_{(\phi)}$, and then $g' = p' \circ g''$, where $g''$ is an $S''$-morphism $T \to (X_{(\phi)})_{(\phi')}$:

$$\begin{tikzcd}
X \arrow[d, "\pi"'] & X_{(\phi)} \arrow[l, "p"'] \arrow[d, "\pi'"'] & (X_{(\phi)})_{(\phi')} \arrow[l, "p'"'] \arrow[d, "\pi''"] \\
S & S' \arrow[l, "\phi"] & S'' \arrow[l, "\phi'"]
\end{tikzcd}$$

So the result follows by the uniqueness of the solution to a universal mapping problem. □ **I** | 110

This result can be written as the equality (up to a canonical isomorphism) $(X_{(S')})_{(S'')} = X_{(S'')}$ (if there is no chance of confusion), or also as

$$(X \times_S S') \times_{S'} S'' = X \times_S S''; \tag{3.3.9.1}$$

the functorial nature of the isomorphism defined in (3.3.9) can similarly be expressed by the transitivity formula for base change morphisms

$$(f_{(S')})_{(S'')} = f_{(S'')} \tag{3.3.9.2}$$

for each $S$-morphism $f : X \to Y$.

**Corollary (3.3.10). —** *If $X$ and $Y$ are $S$-preschemes, then there exists a canonical functorial isomorphism from the $S'$-prescheme $X_{(S')} \times_{S'} Y_{(S')}$ to the $S'$-prescheme $(X \times_S Y)_{(S')}$.*

*Proof.* We have, up to canonical isomorphism,

$$(X \times_S S') \times_{S'} (Y \times_S S') = X \times_S (Y \times_S S') = (X \times_S Y) \times_S S'$$

according to (3.3.9.1) and the associativity of products of $S$-preschemes. □

The functorial nature of the isomorphism defined in Corollary (3.3.10) can be expressed by the formula

$$(u_{(S')}, v_{(S')})_{S'} = ((u,v)_S)_{(S')} \tag{3.3.10.1}$$

for each pair of $S$-morphisms $u : T \to X$, $v : T \to Y$.

In other words, the base change functor $X_{(S')}$ *commutes with products*; it also commutes with sums (3.2.8).

**Corollary (3.3.11). —** *Let $Y$ be an $S$-prescheme, and $f : X \to Y$ a morphism which makes $X$ a $Y$-prescheme (and, as a result, also an $S$-prescheme). The prescheme $X_{(S')}$ is then identified with the product $X \times_Y Y_{(S')}$, the projection $X \times_Y Y_{(S')} \to Y_{(S')}$ being identified with $f_{(S')}$.*

*Proof.* Let $\psi : Y \to S$ be the structure morphism of $Y$; we have the commutative diagram

$$\begin{tikzcd}
S' \arrow[d] & Y_{(S')} \arrow[l, "\psi_{(S')}"'] \arrow[d] & X_{(S')} \arrow[l, "f_{(S')}"'] \arrow[d] \\
S & Y \arrow[l, "\psi"] & X \arrow[l, "f"]
\end{tikzcd}$$

[3] *[Trans.] The French text prints $f$ here; the defining pair is $(g, \psi)$.*

We have that $Y_{(S')}$ is identified with $S'_{(\psi)}$, and $X_{(S')}$ with $S'_{(\psi\circ f)}$; taking (3.3.9) and (3.3.4) into account, we thus deduce the corollary. □

**(3.3.12).** Let $f : X \to X'$ and $g : Y \to Y'$ be $S$-morphisms which are *monomorphisms* of preschemes (T, I, 1.1); then $f \times_S g$ is a *monomorphism*. Indeed, if $p$ and $q$ are the projections of $X \times_S Y$, $p'$ and $q'$ the projections of $X' \times_S Y'$, and $u$ and $v$ are $S$-morphisms $T \to X \times_S Y$, then the equation $(f \times_S g) \circ u = (f \times_S g) \circ v$ implies that $p' \circ (f \times_S g) \circ u = p' \circ (f \times_S g) \circ v$, or, in other words, that $f \circ p \circ u = f \circ p \circ v$, and since $f$ is a monomorphism, $p \circ u = p \circ v$; using the fact that $g$ is a monomorphism, we similarly obtain $q \circ u = q \circ v$, hence $u = v$.

It follows that, for each base change $S' \to S$, **I** | 111

$$f_{(S')} : X_{(S')} \longrightarrow X'_{(S')}$$

is a monomorphism.[4]

**(3.3.13).** Let $S$ and $S'$ be affine schemes of rings $A$ and $A'$ (respectively); a morphism $S' \to S$ then corresponds to a ring homomorphism $A \to A'$. If $X$ is an $S$-prescheme, we denote by $X_{(A')}$ or $X \otimes_A A'$ the $S'$-prescheme $X_{(S')}$; when $X$ is also affine of ring $B$, $X_{(A')}$ is affine of ring $B_{(A')} = B \otimes_A A'$ obtained by extension of scalars from the $A$-algebra $B$ to $A'$.

**(3.3.14).** With the notation of (3.3.6), for each *$S$-morphism* $f : S' \to X$, we have that $f' = (f, 1_{S'})_S$ is an $S'$-morphism $S' \to X' = X_{(S')}$ such that $p \circ f' = f$, $\pi' \circ f' = 1_{S'}$, or, in other words, an *$S'$-section of $X'$*; conversely, if $f'$ is such an $S'$-section, then $f = p \circ f'$ is an $S$-morphism $S' \to X$. We thus define a canonical *bijective correspondence*

$$\mathrm{Hom}_S(S', X) \simeq \mathrm{Hom}_{S'}(S', X').$$

We say that $f'$ is the *graph morphism* of $f$, and we denote it by $\Gamma_f$.

**(3.3.15).** Given a prescheme $X$, which we can always consider as a $\mathbf{Z}$-prescheme, it follows, in particular, from (3.3.14) that the *$X$-sections* of $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ (where $T$ is an indeterminate) correspond bijectively with *morphisms* $X \to \mathbf{Z}[T]$[5]. We will show that these $X$-sections also correspond bijectively with *sections of the structure sheaf $\mathscr{O}_X$ over $X$*. Indeed, let $(U_\alpha)$ be a cover of $X$ by affine open subsets; let $u : X \to X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ be an $X$-morphism, and let $u_\alpha$ be its restriction to $U_\alpha$; if $A_\alpha$ is the ring of the affine scheme $U_\alpha$, then $U_\alpha \otimes_{\mathbf{Z}} \mathbf{Z}[T]$ is an affine scheme of ring $A_\alpha[T]$ (3.2.2), and $u_\alpha$ canonically corresponds to an $A_\alpha$-homomorphism $A_\alpha[T] \to A_\alpha$ (1.7.3). Now, since such a homomorphism is completely determined by the data of the image of $T$ in $A_\alpha$, let $s_\alpha \in A_\alpha = \Gamma(U_\alpha, \mathscr{O}_X)$, and if we suppose that the restrictions of $u_\alpha$ and $u_\beta$ to an affine open subset $V \subset U_\alpha \cap U_\beta$ coincide, then we see immediately that $s_\alpha$ and $s_\beta$ coincide on $V$; thus the family $(s_\alpha)$ consists of the restrictions to $U_\alpha$ of a section $s$ of $\mathscr{O}_X$ over $X$; conversely, it is clear that such a section defines a family $(u_\alpha)$ of morphisms which are the restrictions to $U_\alpha$ of an $X$-morphism $X \to X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$. This result is generalized in (**II**, 1.7.12).

[4] *[Trans.] The French formula prints $Y_{(S')}$; since 3.3.12 assumes $f : X \to X'$, the base-changed codomain is $X'_{(S')}$.*

[5] *[Trans.] The French text prints $\mathbf{Z}[T] \to X$; Proposition (3.3.14) and the ensuing local ring homomorphisms $A_\alpha[T] \to A_\alpha$ require the scheme morphism $X \to \mathbf{Z}[T]$.*

### 3.4. Points of a prescheme with values in a prescheme; geometric points

**(3.4.1).** Let $X$ be a prescheme; for each prescheme $T$, we then denote by $X(T)$ the set $\operatorname{Hom}(T, X)$ of morphisms $T \to X$, and the elements of this set are called *the points of $X$ with values in $T$*. If we associate to each morphism $f : T \to T'$ the map $u' \mapsto u' \circ f$ from $X(T')$ to $X(T)$, we see, for fixed $X$, that $X(T)$ is a *contravariant functor in $T$*, from the category of preschemes to that of sets. In addition, each morphism of preschemes $g : X \to Y$ defines a functorial homomorphism $X(T) \to Y(T)$, which sends $v \in X(T)$ to $g \circ v$.

**(3.4.2).** Given sets $P$, $Q$, and $R$, and maps $\phi : P \to R$ and $\psi : Q \to R$, we define the *fibre product of $P$ and $Q$ over $R$* (with respect to $\phi$ and $\psi$) as the subset of the product set $P \times Q$ consisting of the pairs $(p, q)$ such that $\phi(p) = \psi(q)$; we denote it by $P \times_R Q$. Definition (3.2.1) of the product of $S$-preschemes can be interpreted, with the notation of (3.4.1), via the formula I | 112

(3.4.2.1) $$(X \times_S Y)(T) = X(T) \times_{S(T)} Y(T),$$

where the maps $X(T) \to S(T)$ and $Y(T) \to S(T)$ correspond to the structure morphisms $X \to S$ and $Y \to S$.

**(3.4.3).** If we are given a prescheme $S$ and we consider only the $S$-preschemes and $S$-morphisms, then we will denote by $X(T)_S$ the set $\operatorname{Hom}_S(T, X)$ of $S$-morphisms $T \to X$, and suppress the subscript $S$ when there is no chance of confusion; we say that the elements of $X(T)_S$ are the *points* (or *$S$-points*, when there is a possibility of confusion) *of the $S$-prescheme $X$ with values in the $S$-prescheme $T$*. In particular, an *$S$-section* of $X$ is none other than a *point of $X$ with values in $S$*. The formula (3.4.2.1) can then be written as

(3.4.3.1) $$(X \times_S Y)(T)_S = X(T)_S \times Y(T)_S;$$

more generally, if $Z$ is an $S$-prescheme, and $X$, $Y$, and $T$ are $Z$-preschemes (thus *ipso facto* $S$-preschemes), then we have

(3.4.3.2) $$(X \times_Z Y)(T)_S = X(T)_S \times_{Z(T)_S} Y(T)_S.$$

We note that, to show that a triple $(W, r, s)$ consisting of an $S$-prescheme $W$ and $S$-morphisms $r : W \to X$ and $s : W \to Y$ is a product of $X$ and $Y$ (over $Z$), it suffices, by definition, to check that, for *each $S$-prescheme $T$*, the diagram

$$\begin{tikzcd} W(T)_S \arrow[r, "r'"] \arrow[d, "s'"'] & X(T)_S \arrow[d, "\phi'"] \\ Y(T)_S \arrow[r, "\psi'"'] & Z(T)_S \end{tikzcd}$$

makes $W(T)_S$ the fibre product of $X(T)_S$ and $Y(T)_S$ over $Z(T)_S$, where $r'$ and $s'$ correspond to $r$ and $s$, and $\phi'$ and $\psi'$ to the structure morphisms $\phi : X \to Z$ and $\psi : Y \to Z$.

**(3.4.4).** When $T$ (resp. $S$) in the above is an affine scheme of ring $B$ (resp. $A$), we replace $T$ (resp. $S$) by $B$ (resp. $A$) in the above notation, and we then call the elements of $X(B)$ the *points of $X$ with values in the ring $B$*, and the elements of $X(B)_A$ the *points of the $A$-prescheme $X$ with values in the $A$-algebra $B$*. We note that $X(B)$ and $X(B)_A$ are *covariant* functors in $B$. We similarly write $X(T)_A$ for the set of points of the $A$-prescheme $X$ with values in the $A$-prescheme $T$.

**(3.4.5).** Consider, in particular, the case where $T$ is of the form $\operatorname{Spec}(A)$, where $A$ is a *local* ring; the elements of $X(A)$ then correspond bijectively to *local* homomorphisms $\mathscr{O}_x \to A$ for $x \in X$ (2.4.4); we say that the point $x$ of the underlying space of $X$ is the *location*[6] of the point of $X$ with values in $A$ to which it corresponds.

More specifically, we define the *geometric points* of a prescheme $X$ to be the *points of $X$ with values in a field $K$*: the data of such a point is equivalent to the data of its location $x$ in the underlying space of $X$, and of an *extension* $K$ of $k(x)$; $K$ will be called the *field of values* of the corresponding geometric I | 113

[6] *[Trans.] We also say that* the geometric point lies over this $x$.

point, and we say that this geometric point is *located at* $x$. We thus define a map $X(K) \to X$, sending a geometric point with values in $K$ to its location.

If $S' = \operatorname{Spec}(K)$ is an $S$-prescheme (in other words, if $K$ is considered as an extension of the residue field $k(s)$, where $s \in S$), and if $X$ is an $S$-prescheme, then an element of $X(K)_S$, or, as we say, a *geometric point of $X$ lying over $s$ with values in $K$*, consists of the data of a $k(s)$-monomorphism from the residue field $k(x)$ to $K$, where $x$ is a point of $X$ *lying over* $s$ (therefore $k(x)$ is an extension of $k(s)$).

In particular, if $S = \operatorname{Spec}(K) = \{\xi\}$, *then the geometric points of $X$ with values in $K$ can be identified with the points $x \in X$ such that $k(x) = K$*; we say that these latter points are the *$K$-rational points of the $K$-prescheme $X$*; if $K'$ is an extension of $K$, then the geometric points of $X$ with values in $K'$ bijectively correspond to the $K'$-rational points of $X' = X_{(K')}$ (3.3.14).

**Lemma (3.4.6).** — *Let $X_i$ ($1 \leqslant i \leqslant n$) be $S$-preschemes, $s$ a point of $S$, and $x_i$ ($1 \leqslant i \leqslant n$) points of $X_i$ lying over $s$. Then there exists an extension $K$ of $k(s)$ and a geometric point of the product $Y = X_1 \times_S X_2 \times_S \cdots \times_S X_n$, with values in $K$, whose projections to the $X_i$ are localized at the $x_i$.*

*Proof.* There exist $k(s)$-monomorphisms $k(x_i) \to K$, all in the same extension $K$ of $k(s)$ (Bourbaki, *Alg.*, chap. V, §4, prop. 2). The compositions $k(s) \to k(x_i) \to K$ are all identical, and so the morphisms $\operatorname{Spec}(K) \to X_i$ corresponding to the $k(x_i) \to K$ are all $S$-morphisms, and we thus conclude that they define a unique morphism $\operatorname{Spec}(K) \to Y$. If $y$ is the corresponding point of $Y$, it is clear that its projection to each of the $X_i$ is $x_i$. □

**Proposition (3.4.7).** — *Let $X_i$ ($1 \leqslant i \leqslant n$) be $S$-preschemes, and, for each index $i$, let $x_i$ be a point of $X_i$. For there to exist a point $y$ of $Y = X_1 \times_S X_2 \times \ldots \times_S X_n$ whose image is $x_i$ under the $i$th projection for each $1 \leqslant i \leqslant n$, it is necessary and sufficient that the $x_i$ all lie above the same point $s$ of $S$.*

*Proof.* The condition is evidently necessary; Lemma (3.4.6) proves that it is sufficient. □

In other words, if we denote by $(X)$ the underlying set of $X$, we see that we have a canonical *surjective* function $(X \times_S Y) \to (X) \times_{(S)} (Y)$; we must point out that this function *is not injective* in general; in other words, *there can exist multiple distinct points $z$ in $X \times_S Y$ that have the same projections $x \in X$ and $y \in Y$*; we have already seen this when $S$, $X$, and $Y$ are prime spectra of fields $k$, $K$, and $K'$ (respectively), since the tensor product $K \otimes_k K'$ has, in general, multiple distinct prime ideals (cf. (3.4.9)).

**Corollary (3.4.8).** — *Let $f : X \to Y$ be an $S$-morphism, and $f_{(S')} : X_{(S')} \to Y_{(S')}$ the $S'$-morphism obtained from $f$ by the base extension $S' \to S$. Let $p$ (resp. $q$) be the projection $X_{(S')} \to X$ (resp. $Y_{(S')} \to Y$); for every subset $M$ of $X$, we have*

$$q^{-1}(f(M)) = f_{(S')}(p^{-1}(M)).$$

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*Proof.* Indeed, by (3.3.11), $X_{(S')}$ can be identified with the product $X \times_Y Y_{(S')}$ thanks to the commutative diagram

$$\begin{tikzcd} X & X_{(S')} \arrow[l, "p"'] \arrow[d, "f_{(S')}"] \\ Y \arrow[from=u, "f"'] & Y_{(S')} \arrow[l, "q"] \end{tikzcd}$$

By (3.4.7), the equation $q(y') = f(x)$ for $x \in M$ and $y' \in Y_{(S')}$ is equivalent to the existence of some $x' \in X_{(S')}$ such that $p(x') = x$ and $f_{(S')}(x') = y'$, whence the corollary. □

Lemma (3.4.6) can be made clearer in the following manner:

**Proposition (3.4.9).** — *Let $X$ and $Y$ be $S$-preschemes, $x$ a point of $X$, and $y$ a point of $Y$, with both $x$ and $y$ lying above the same point $s \in S$. The set of points of $X \times_S Y$ with projections $x$ and $y$ is in canonical bijective correspondence with the set of types of composite extensions of $k(x)$ and $k(y)$ considered as extensions of $k(s)$* (Bourbaki, *Alg.*, chap. VIII, §8, prop. 2).

*Proof.* Let $p$ (resp. $q$) be the projection from $X \times_S Y$ to $X$ (resp. $Y$), and $E$ the subspace $p^{-1}(x) \cap q^{-1}(y)$ of the underlying space of $X \times_S Y$. First, note that the morphisms $\mathrm{Spec}(k(x)) \to S$ and $\mathrm{Spec}(k(y)) \to S$ factor as $\mathrm{Spec}(k(x)) \to \mathrm{Spec}(k(s)) \to S$ and $\mathrm{Spec}(k(y)) \to \mathrm{Spec}(k(s)) \to S$; since $\mathrm{Spec}(k(s)) \to S$ is a monomorphism (2.4.7), it follows from (3.2.4) that we have

$$P = \mathrm{Spec}(k(x)) \times_S \mathrm{Spec}(k(y)) = \mathrm{Spec}(k(x)) \times_{\mathrm{Spec}(k(s))} \mathrm{Spec}(k(y)) = \mathrm{Spec}(k(x) \otimes_{k(s)} k(y)).$$

We will define two maps, $\alpha : P_0 \to E$ and $\beta : E \to P_0$, inverse to one another (where $P_0$ denotes the underlying set of the prescheme $P$). If $i : \mathrm{Spec}(k(x)) \to X$ and $j : \mathrm{Spec}(k(y)) \to Y$ are the canonical morphisms (2.4.5), we take $\alpha$ to be the map of underlying spaces corresponding to the morphism $i \times_S j$. On the other hand, every $z \in E$ defines, by hypothesis, two $k(s)$-monomorphisms, $k(x) \to k(z)$ and $k(y) \to k(z)$, and thus a $k(s)$-homomorphism[7] $k(x) \otimes_{k(s)} k(y) \to k(z)$, and thus a morphism $\mathrm{Spec}(k(z)) \to P$; $\beta(z)$ will be the image of $z$ in $P_0$ under this morphism. The verification of the fact that $\alpha \circ \beta$ and $\beta \circ \alpha$ are the identity maps follows from (2.4.5) and the definition of the product (3.2.1). Finally, we know that $P_0$ is in bijective correspondence with the set of types of composite extensions of $k(x)$ and $k(y)$ (Bourbaki, *Alg.*, chap. VIII, §8, prop. 1). □

### 3.5. Surjections and injections

**(3.5.1).** In a general sense, consider a property **P** of morphisms of preschemes, and the following two propositions:

(i) If $f : X \to X'$ and $g : Y \to Y'$ are $S$-morphisms that have property **P**, then $f \times_S g$ also has property **P**.

(ii) If $f : X \to Y$ is an $S$-morphism that has property **P**, then every $S'$-morphism $f_{(S')} : X_{(S')} \to Y_{(S')}$ obtained from $f$ by a base extension $S' \to S$ also has property **P**.

Since $f_{(S')} = f \times_S 1_{S'}$, we see that, if, for every prescheme $X$, the *identity* $1_X$ has property **P**, then (i) implies (ii); on the other hand, since $f \times_S g$ is the composite morphism

$$X \times_S Y \xrightarrow{f \times 1_Y} X' \times_S Y \xrightarrow{1_{X'} \times g} X' \times_S Y',$$

we see that, if the *composite* of two morphisms having property **P** also has property **P**, then so does the product $f \times_S g$, and so (ii) implies (i).

A first application of this remark is

**Proposition (3.5.2). —**

(i) *If $f : X \to X'$ and $g : Y \to Y'$ are surjective $S$-morphisms, then $f \times_S g$ is surjective.*

(ii) *If $f : X \to Y$ is a surjective $S$-morphism, then $f_{(S')}$ is surjective for every base extension $S' \to S$.*

*Proof.* The composition of any two surjections being a surjection, it suffices to prove (ii); but this proposition follows immediately from (3.4.8) applied to $M = X$. □

**Proposition (3.5.3). —** *For a morphism $f : X \to Y$ to be surjective, it is necessary and sufficient that, for every field $K$ and every morphism $\mathrm{Spec}(K) \to Y$, there exist an extension $K'$ of $K$ and a morphism $\mathrm{Spec}(K') \to X$ that make the following diagram commute:*

$$\begin{tikzcd} X \arrow[d, "f"'] & \mathrm{Spec}(K') \arrow[l] \arrow[d] \\ Y & \mathrm{Spec}(K). \arrow[l] \end{tikzcd}$$

*Proof.* The condition is sufficient because, for all $y \in Y$, it suffices to apply it to a morphism $\mathrm{Spec}(K) \to Y$ corresponding to a monomorphism $k(y) \to K$, with $K$ being an extension of $k(y)$ (2.4.6). Conversely, suppose that $f$ is surjective, and let $y \in Y$ be the image of the unique point of $\mathrm{Spec}(K)$; there exists some $x \in X$ such that $f(x) = y$; we will consider the corresponding monomorphism $k(y) \to k(x)$ (2.2.1); it then suffices to take $K'$ to be an extension of $k(y)$ such that there exist

[7] *[Trans.] The French text prints "monomorphism" here. The induced map from the tensor product need not be injective; only a homomorphism is needed to define the ensuing morphism of spectra.*

$k(y)$-monomorphisms from $k(x)$ and $K$ to $K'$ (Bourbaki, *Alg.*, chap. V, ğ4, prop. 2); the morphism $\operatorname{Spec}(K') \to X$ corresponding to $k(x) \to K'$ is exactly that for which we are searching. □

With the language introduced in (3.4.5), we can further say that *every geometric point of $Y$ with values in $K$ comes from a geometric point of $X$ with values in an extension of $K$.*

**Definition (3.5.4). —** We say that a morphism $f : X \to Y$ of preschemes is *universally injective*, or a *radicial morphism*, if, for every field $K$, the corresponding map $X(K) \to Y(K)$ is injective.

It follows immediately from the definitions that every *monomorphism of preschemes* (T, 1.1) is radicial.

**(3.5.5).** For a morphism $f : X \to Y$ to be radicial, it suffices that the condition of Definition (3.5.4) hold for every *algebraically closed* field. In fact, if $K$ is an arbitrary field, and $K'$ an algebraically closed extension of $K$, then the diagram

$$\begin{tikzcd} X(K) \arrow[r, "\alpha"] \arrow[d, "\phi"'] & Y(K) \arrow[d, "\phi'"] \\ X(K') \arrow[r, "\alpha'"'] & Y(K') \end{tikzcd}$$

commutes, where $\phi$ and $\phi'$ come from the morphism $\operatorname{Spec}(K') \to \operatorname{Spec}(K)$, and $\alpha$ and $\alpha'$ correspond to $f$. However, $\phi$ is injective, and so too is $\alpha'$, by hypothesis; hence $\alpha$ is necessarily injective.

**Proposition (3.5.6). —** *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of preschemes.*

(i) *If $f$ and $g$ are radicial, then so is $g \circ f$.*
(ii) *Conversely, if $g \circ f$ is radicial, then so is $f$.*

*Proof.* Taking into account Definition (3.5.4), the proposition reduces to the corresponding claims for the maps $X(K) \to Y(K) \to Z(K)$, and these claims are evident. □

**Proposition (3.5.7). —**

(i) *If the $S$-morphisms $f : X \to X'$ and $g : Y \to Y'$ are radicial, then so is $f \times_S g$.*
(ii) *If the $S$-morphism $f : X \to Y$ is radicial, then so is $f_{(S')} : X_{(S')} \to Y_{(S')}$ for every extension $S' \to S$ of the base prescheme.*

*Proof.* Given (3.5.1), it suffices to prove (i). We have seen in (3.4.2.1) that

$$(X \times_S Y)(K) = X(K) \times_{S(K)} Y(K),$$
$$(X' \times_S Y')(K) = X'(K) \times_{S(K)} Y'(K),$$

with the map $(X \times_S Y)(K) \to (X' \times_S Y')(K)$ corresponding to $f \times_S g$ thus being identified with $(u, v) \to (f \circ u, g \circ v)$, and the proposition follows immediately. □

**Proposition (3.5.8). —** *For a morphism $f = (\psi, \theta) : X \to Y$ to be radicial, it is necessary and sufficient for $\psi$ to be injective and for the monomorphism $\theta^x : k(\psi(x)) \to k(x)$ to make $k(x)$ a radicial extension of $k(\psi(x))$ for every $x \in X$.*

*Proof.* We suppose that $f$ is radicial and first show that the equation $\psi(x_1) = \psi(x_2) = y$ necessarily implies that $x_1 = x_2$. Indeed, there exists a field $K$ extending $k(y)$, together with $k(y)$-monomorphisms $k(x_1) \to K$ and $k(x_2) \to K$ (Bourbaki, *Alg.*, chap. V, ğ4, prop. 2); the corresponding morphisms $u_1 : \operatorname{Spec}(K) \to X$ and $u_2 : \operatorname{Spec}(K) \to X$ are then such that $f \circ u_1 = f \circ u_2$, and so $u_1 = u_2$ by hypothesis, and this implies, in particular, that $x_1 = x_2$. We now consider $k(x)$ as an extension of $k(\psi(x))$ by means of $\theta^x$: if $k(x)$ is not a radicial extension of $k(\psi(x))$, then there exist two distinct $k(\psi(x))$-monomorphisms from $k(x)$ to an algebraically closed extension $K$ of $k(\psi(x))$, and the two corresponding morphisms $\operatorname{Spec}(K) \to X$ would contradict the hypothesis. Conversely, taking (2.4.6) into account, it is immediate that the conditions stated are sufficient for $f$ to be radicial. □

**Corollary (3.5.9). —** *If $A$ is a ring, and $S$ is a multiplicative set of $A$, then the canonical morphism $\operatorname{Spec}(S^{-1}A) \to \operatorname{Spec}(A)$ is radicial.*

*Proof.* Indeed, this morphism is a monomorphism (1.6.2). □

**Corollary (3.5.10).** — *Let $f : X \to Y$ be a radicial morphism, $g : Y' \to Y$ a morphism, and $X' = X_{(Y')} = X \times_Y Y'$. Then the radicial morphism $f_{(Y')}$ (3.5.7, ii) is a bijection from the underlying space of $X'$ to $g^{-1}(f(X))$; further, for every field $K$, the set $X'(K)$ can be identified with the inverse image of the subset $X(K) \subseteq Y(K)$ under the map $Y'(K) \to Y(K)$ corresponding to $g$.*

*Proof.* The first claim follows immediately from (3.5.8) and (3.4.8); the second, from the commutativity of the following diagram: I | 117

$$\begin{tikzcd} X'(K) \arrow[r] \arrow[d] & Y'(K) \arrow[d] \\ X(K) \arrow[r] & Y(K) \end{tikzcd}$$

□

**Remark (3.5.11).** — We say that a morphism $f = (\psi, \theta)$ of preschemes is *injective* if the map $\psi$ is injective. For a morphism $f = (\psi, \theta) : X \to Y$ to be radicial, it is necessary and sufficient that, for every morphism $Y' \to Y$, the morphism $f_{(Y')} : X_{(Y')} \to Y'$ be injective (which justifies the terminology of a *universally injective* morphism). In fact, the condition is necessary by (3.5.7, ii) and (3.5.8). Conversely, the condition first implies that $\psi$ is injective; if, for some $x \in X$, the monomorphism $\theta^x : k(\psi(x)) \to k(x)$ were not radicial, then there would be an extension $K$ of $k(\psi(x))$, and two distinct morphisms $\operatorname{Spec}(K) \to X$ corresponding to the same morphism $\operatorname{Spec}(K) \to Y$ (3.5.8). But then, setting $Y' = \operatorname{Spec}(K)$, there would be two distinct $Y'$-sections of $X_{(Y')}$ (3.3.14), which contradicts the hypothesis that $f_{(Y')}$ is injective.

### 3.6. Fibres

**Proposition (3.6.1).** — *Let $f : X \to Y$ be a morphism, $y$ a point of $Y$, and $\mathfrak{a}_y$ an ideal of definition of $\mathscr{O}_y$ for the $\mathfrak{m}_y$-preadic topology. Then the projection $p : X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to X$ is a homeomorphism from the underlying space of the prescheme $X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ to the fibre $f^{-1}(y)$ equipped with the topology induced from that of the underlying space of $X$.*

*Proof.* Since $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y) \to Y$ is radicial ((3.5.4) and (2.4.7)), since $\operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ is a single point, and since the ideal $\mathfrak{m}_y/\mathfrak{a}_y$ is nilpotent by hypothesis (1.1.12), we already know ((3.5.10) and (3.3.4)) that $p$ identifies, as *sets*, the underlying space of $X \times_Y \operatorname{Spec}(\mathscr{O}_y/\mathfrak{a}_y)$ with $f^{-1}(y)$; everything reduces to proving that $p$ is a homeomorphism. By (3.2.7), the question is local on $X$ and $Y$, and so we can suppose that $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$, with $B$ being an $A$-algebra. The morphism $p$ then corresponds to the homomorphism $1 \otimes \phi : B \to B \otimes_A A'$, where $A' = A_y/\mathfrak{a}_y$ and $\phi$ is the canonical map from $A$ to $A'$. Then every element of $B \otimes_A A'$ can be written as

$$\sum_i b_i \otimes \phi(a_i)/\phi(s) = \left(\sum_i (a_i b_i \otimes 1)\right)(1 \otimes \phi(s))^{-1},$$

where $s \notin \mathfrak{j}_y$, and Proposition (1.2.4) applies. □

**(3.6.2).** Throughout the rest of this treatise, whenever we consider a fibre $f^{-1}(y)$ of a morphism as having the structure of a $k(y)$-prescheme, *it will always be the prescheme obtained by transporting the structure of $X \times_Y \operatorname{Spec}(k(y))$ by the projection to $X$*. We will also write this latter product as $X \times_Y k(y)$, or $X \otimes_{\mathscr{O}_Y} k(y)$; more generally, if $B$ is an $\mathscr{O}_y$-algebra, we will denote by $X \times_Y B$ or $X \otimes_{\mathscr{O}_Y} B$ the product $X \times_Y \operatorname{Spec}(B)$.

With the preceding convention, it follows from (3.5.10) that the points of $X$ with values in an extension $K$ of $k(y)$ are identified with the *points of $f^{-1}(y)$ with values in $K$*.

**(3.6.3).** Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms, and $h = g \circ f$ their composition; for all $z \in Z$, the fibre $h^{-1}(z)$ is a prescheme isomorphic to

$$X \times_Z \operatorname{Spec}(k(z)) = (X \times_Y Y) \times_Z \operatorname{Spec}(k(z)) = X \times_Y g^{-1}(z).$$

In particular, if $U$ is an open subset of $X$, then the prescheme induced on $U \cap f^{-1}(y)$ by the prescheme $f^{-1}(y)$ is isomorphic to $f_U^{-1}(y)$ ($f_U$ being the restriction of $f$ to $U$). I | 118

**Proposition (3.6.4).** — (Transitivity of fibres) *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms; let $X' = X \times_Y Y' = X_{(Y')}$ and $f' = f_{(Y')} : X' \to Y'$. For every $y' \in Y'$, if we let $y = g(y')$, then the prescheme $f'^{-1}(y')$ is isomorphic to $f^{-1}(y) \otimes_{k(y)} k(y')$.*

*Proof.* Indeed, it suffices to remark that the two preschemes $(X \otimes_Y k(y)) \otimes_{k(y)} k(y')$ and $(X \times_Y Y') \otimes_{Y'} k(y')$ are both canonically isomorphic to $X \times_Y \operatorname{Spec}(k(y'))$ by (3.3.9.1). □

In particular, if $V$ is an open neighbourhood of $y$ in $Y$, and we denote by $f_V$ the restriction of $f$ to the induced prescheme on $f^{-1}(V)$, then the preschemes $f^{-1}(y)$ and $f_V^{-1}(y)$ are canonically identified.

**Proposition (3.6.5).** — *Let $f : X \to Y$ be a morphism, $y$ a point of $Y$, $Z$ the local prescheme* $\operatorname{Spec}(\mathscr{O}_y)$*, and $p = (\psi, \theta)$ the projection $X \times_Y Z \to X$; then $p$ is a homeomorphism from the underlying space of $X \times_Y Z$ to the subspace $f^{-1}(Z)$ of $X$ (*when the underlying space of $Z$ is identified with a subspace of $Y$, cf. (2.4.2)*), and, for all $t \in X \times_Y Z$, letting $z = \psi(t)$, we have that $\theta_t^\sharp$ is an isomorphism from $\mathscr{O}_z$ to $\mathscr{O}_t$.*

*Proof.* Since $Z$ (identified as a subspace of $Y$) is contained in every affine open containing $y$ (2.4.2), we can, as in (3.6.1), reduce to the case where $X = \operatorname{Spec}(A)$ and $Y = \operatorname{Spec}(B)$ are affine schemes, with $A$ being a $B$-algebra. Then $X \times_Y Z$ is the prime spectrum of $A \otimes_B B_y$, and this ring is canonically identified with $S^{-1}A$, where $S$ is the image of $B - \mathfrak{j}_y$ in $A$ (**0**, 1.5.2); since $p$ then corresponds to the canonical homomorphism $A \to S^{-1}A$, the proposition follows from (1.6.2). □

**3.7. Application: reduction of a prescheme mod.** $\mathfrak{J}$ *This section, which makes use of notions and results from later in Chapter I and from Chapter II, will not be used in what follows in this treatise, and is only intended for readers familiar with classical algebraic geometry.*

**(3.7.1).** Let $A$ be a ring, $X$ an $A$-prescheme, and $\mathfrak{J}$ an ideal of $A$; then $X_0 = X \otimes_A (A/\mathfrak{J})$ is an $(A/\mathfrak{J})$-prescheme, which we sometimes say is induced from $X$ by *reduction* mod. $\mathfrak{J}$.

**(3.7.2).** This terminology is used especially when $A$ is a *local ring* and $\mathfrak{J}$ its maximal ideal, so that $X_0$ is a prescheme over the residue field $k = A/\mathfrak{J}$ of $A$.

When $A$ is also integral, with field of fractions $K$, we can consider the $K$-prescheme $X' = X \otimes_A K$. By an abuse of language which we will not use, it has been customary until now to say that $X_0$ is *induced from* $X'$ by reduction mod. $\mathfrak{J}$. In the cases where this language was used, $A$ was a local ring of dimension 1 (most often a discrete valuation ring), and it was understood (more or less explicitly) that the given $K$-prescheme $X'$ was a closed subprescheme of a $K$-prescheme $P'$ (in fact, a projective space of type $\mathbf{P}_K^r$, cf. (**II**, 4.1.1)), itself of the form $P' = P \otimes_A K$, where $P$ is a given $A$-prescheme (in fact, the $A$-scheme $\mathbf{P}_A^r$, with the notation of (**II**, 4.1.1)). In our language, the definition of $X_0$ in terms of $X'$ is formulated as follows:

We consider the affine scheme $Y = \operatorname{Spec}(A)$, formed of two points, the unique closed point $y = \mathfrak{J}$ **I** | 119
and the generic point $(0)$, the singleton set $U$ of the generic point being thus an open $U = \operatorname{Spec}(K)$ in $Y$. If $X$ is an $A$-prescheme (or, in other words, a $Y$-prescheme), then $X \otimes_A K = X'$ is exactly the prescheme induced by $X$ on $\psi^{-1}(U)$, denoting by $\psi$ the structure morphism $X \to Y$. In particular, if $\phi$ is the structure morphism $P \to Y$, then a closed subprescheme $X'$ of $P' = \phi^{-1}(U)$ is a (locally closed) subprescheme of $P$. If $P$ is Noetherian (for example, if $A$ is Noetherian and $P$ is of finite type over $A$), then there exists a smallest closed subprescheme $X = \overline{X'}$ of $P$ containing $X'$ (9.5.10), and $X'$ is the prescheme induced by $X$ on the open $\phi^{-1}(U) \cap X$, and so is isomorphic to $X \otimes_A K$ (9.5.10). *The immersion of $X'$ into $P' = P \otimes_A K$ thus lets us canonically consider $X'$ as being of the form $X' = X \otimes_A K$, where $X$ is an $A$-prescheme.* We can then consider the reduced mod. $\mathfrak{J}$ prescheme $X_0 = X \otimes_A k$, which is exactly the fibre $\psi^{-1}(y)$ of the closed point $y$. Up until now, lacking the adequate terminology, we had avoided explicitly introducing the $A$-prescheme $X$. One should note, however, that all the claims normally made about the "reduced mod. $\mathfrak{J}$" prescheme $X_0$ should be seen as consequences of more complete claims about $X$ itself, and cannot be satisfactorily formulated or understood except by interpreting them as such. It seems also that any hypotheses made on $X_0$ always reduce to hypotheses on $X$ itself (independent of the prior data of an immersion of $X'$ in $\mathbf{P}_K^r$), which lets us give more intrinsic statements.

**(3.7.3).** Lastly, we draw attention to a very particular fact, which has undoubtedly contributed to slowing the conceptual clarification of the situation envisaged here: if $A$ is a discrete valuation ring, and if $X$ is *proper* over $A$ (which is indeed the case if $X$ is a closed subprescheme of some $\mathbf{P}^r_A$, cf. (**II**, 5.5.4)), then the points of $X$ with values in $A$ and the points of $X'$ with values in $K$ are in bijective correspondence (**II**, 7.3.8). This is why we have often thought we were proving facts about $X'$, when in reality we were proving facts about $X$; those facts remain valid (in this form) when the base local ring is no longer assumed to have dimension 1.

## §4. SUBPRESCHEMES AND IMMERSION MORPHISMS

### 4.1. Subpreschemes

**(4.1.1).** As the notion of a quasi-coherent sheaf (**0**, 5.1.3) is local, a quasi-coherent $\mathscr{O}_X$-module $\mathscr{F}$ on a prescheme $X$ can be defined by the following condition: for each affine open $V$ of $X$, $\mathscr{F}|V$ is isomorphic to the sheaf associated to a $\Gamma(V, \mathscr{O}_X)$-module (1.4.1). It is clear that, on a prescheme $X$, the structure sheaf $\mathscr{O}_X$ is quasi-coherent, and that the kernels, cokernels, and images of homomorphisms of quasi-coherent $\mathscr{O}_X$-modules, as well as inductive limits and direct sums of quasi-coherent $\mathscr{O}_X$-modules, are also quasi-coherent (Theorem (1.3.7) and Corollary (1.3.9)).

**Proposition (4.1.2).** — *Let $X$ be a prescheme, and $\mathscr{I}$ a quasi-coherent sheaf of ideals of $\mathscr{O}_X$. Then the support $Y$ of the sheaf $\mathscr{O}_X/\mathscr{I}$ is closed, and if we denote by $\mathscr{O}_Y$ the restriction of $\mathscr{O}_X/\mathscr{I}$ to $Y$, then $(Y, \mathscr{O}_Y)$ is a prescheme.*

*Proof.* It evidently suffices (2.1.3) to consider the case where $X$ is an affine scheme, and to show that, in this case, $Y$ is closed in $X$, and is an *affine scheme*. Indeed, if $X = \operatorname{Spec}(A)$, then we have $\mathscr{O}_X = \widetilde{A}$ and $\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is an ideal of $A$ (1.4.1); $Y$ is then equal to the closed subset $V(\mathfrak{I})$ of $X$ and can be identified with the prime spectrum of the ring $B = A/\mathfrak{I}$ (1.1.11); in addition, if $\phi$ is the canonical homomorphism $A \to B = A/\mathfrak{I}$, then the direct image ${}^a\phi_*(\widetilde{B})$ is canonically identified with the sheaf $\widetilde{A}/\widetilde{\mathfrak{I}} = \mathscr{O}_X/\mathscr{I}$ (Proposition (1.6.3) and Corollary (1.3.9)), which finishes the proof. ☐ **I | 120**

We say that $(Y, \mathscr{O}_Y)$ is the *subprescheme* of $(X, \mathscr{O}_X)$ *defined by the sheaf of ideals* $\mathscr{I}$; this is a particular case of the more general notion of a *subprescheme*:

**Definition (4.1.3).** — We say that a ringed space $(Y, \mathscr{O}_Y)$ is a subprescheme of a prescheme $(X, \mathscr{O}_X)$ if:

1st. $Y$ is a locally closed subspace of $X$;
2nd. if $U$ denotes the largest open subset of $X$ containing $Y$ such that $Y$ is closed in $U$ (*equivalently*, the complement in $X$ of the boundary of $Y$ with respect to $\overline{Y}$), then $(Y, \mathscr{O}_Y)$ is a subprescheme of $(U, \mathscr{O}_X|U)$ defined by a quasi-coherent sheaf of ideals of $\mathscr{O}_X|U$.

We say that the subprescheme $(Y, \mathscr{O}_Y)$ of $(X, \mathscr{O}_X)$ is closed if $Y$ is closed in $X$ (in which case $U = X$).

It follows immediately from this definition and Proposition (4.1.2) that the closed subpreschemes of $X$ are in canonical *bijective correspondence* with the quasi-coherent sheaves of ideals $\mathscr{J}$ of $\mathscr{O}_X$, since if two such sheaves $\mathscr{J}$ and $\mathscr{J}'$ have the same (closed) support $Y$, and if the restrictions of $\mathscr{O}_X/\mathscr{J}$ and $\mathscr{O}_X/\mathscr{J}'$ to $Y$ are identical, then we have $\mathscr{J}' = \mathscr{J}$.

**(4.1.4).** Let $(Y, \mathscr{O}_Y)$ be a subprescheme of $X$, $U$ the largest open subset of $X$ such that $Y$ is closed (and thus contained) in $U$, and $V$ an open subset of $X$ contained in $U$; then $V \cap Y$ is closed in $V$. In addition, if $Y$ is defined by the quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X|U$, then $\mathscr{J}|V$ is a quasi-coherent sheaf of ideals of $\mathscr{O}_X|V$, and it is immediate that the prescheme induced by $Y$ on $Y \cap V$ is the closed subprescheme of $V$ defined by the sheaf of ideals $\mathscr{J}|V$. Conversely:

**Proposition (4.1.5).** — *Let $(Y, \mathscr{O}_Y)$ be a ringed space such that $Y$ is a subspace of $X$, and suppose there is a cover $(V_\alpha)$ of $Y$ by open subsets of $X$ such that, for each $\alpha$, $Y \cap V_\alpha$ is closed in $V_\alpha$, and the ringed space $(Y \cap V_\alpha, \mathscr{O}_Y|(Y \cap V_\alpha))$ is a closed subprescheme of the prescheme induced on $V_\alpha$ by $X$. Then $(Y, \mathscr{O}_Y)$ is a subprescheme of $X$.*

*Proof.* The hypotheses imply that $Y$ is locally closed in $X$ and that the largest open $U$ in which $Y$ is closed (and thus contained) contains all the $V_\alpha$; we can thus reduce to the case where $U = X$ and $Y$

is closed in $X$. We then define a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$ by taking $\mathscr{J}|V_\alpha$ to be the sheaf of ideals of $\mathscr{O}_X|V_\alpha$ which defines the closed subprescheme $(Y \cap V_\alpha, \mathscr{O}_Y|(Y \cap V_\alpha))$, and, for each open subset $W$ of $X$ not intersecting $Y$, $\mathscr{J}|W = \mathscr{O}_X|W$. We see immediately, by Definition (4.1.3) and (4.1.4), that there exists a unique sheaf of ideals $\mathscr{J}$ satisfying these conditions, and that it defines the closed subprescheme $(Y, \mathscr{O}_Y)$. □

In particular, the *induced* (by $X$) prescheme on an *open subset* of $X$ is a *subprescheme* of $X$.

**Proposition (4.1.6). —** *A subprescheme (resp. a closed subprescheme) of a subprescheme (resp. closed subprescheme) of $X$ is canonically identified with a subprescheme (resp. closed subprescheme) of $X$.* I | 121

*Proof.* Since a locally closed subset of a locally closed subspace of $X$ is a locally closed subspace of $X$, it is clear (4.1.5) that the question is local and that we can thus suppose that $X$ is affine; the proposition then follows from the canonical identification of $A/\mathfrak{J}'$ with $(A/\mathfrak{J})/(\mathfrak{J}'/\mathfrak{J})$, where $A$ is some ring, and $\mathfrak{J}$ and $\mathfrak{J}'$ are ideals of $A$ such that $\mathfrak{J} \subset \mathfrak{J}'$. □

In what follows, we will always make use of the above identification.

**(4.1.7).** Let $Y$ be a subprescheme of a prescheme $X$, and denote by $\psi$ the canonical injection $Y \to X$ of the *underlying subspaces*; we know that the inverse image $\psi^*(\mathscr{O}_X)$ is the restriction $\mathscr{O}_X|Y$ (**0**, 3.7.1). If, for each $y \in Y$, we denote by $\omega_y$ the canonical homomorphism $(\mathscr{O}_X)_y \to (\mathscr{O}_Y)_y$, then these homomorphisms are the restrictions to stalks of a *surjective* homomorphism $\omega$ of sheaves of rings $\mathscr{O}_X|Y \to \mathscr{O}_Y$: indeed, it suffices to check locally on $Y$, that is to say, we can suppose that $X$ is affine and that the subprescheme $Y$ is closed; if in this case $\mathscr{I}$ is the sheaf of ideals of $\mathscr{O}_X$ which defines $Y$, then the $\omega_y$ are none other than the restrictions to stalks of the homomorphism $\mathscr{O}_X|Y \to (\mathscr{O}_X/\mathscr{I})|Y$. We have thus defined a *monomorphism of ringed spaces* (**0**, 4.1.1) $j = (\psi, \omega^\flat)$ which is evidently a morphism $Y \to X$ of preschemes (2.2.1), and we call this the *canonical injection morphism*.

If $f : X \to Z$ is a morphism, we then say that the composite morphism $Y \xrightarrow{j} X \xrightarrow{f} Z$ is the *restriction* of $f$ to the subprescheme $Y$.

**(4.1.8).** Conforming to the general definitions (T, I, 1.1), we will say that a morphism of preschemes $f : Z \to X$ is *majoré*[8] by the injection morphism $j : Y \to X$ of a subprescheme $Y$ of $X$ if $f$ factors as $Z \xrightarrow{g} Y \xrightarrow{j} X$, where $g$ is a morphism of preschemes; further, $g$ is necessarily *unique* since $j$ is a monomorphism.

**Proposition (4.1.9). —** *For a morphism $f : Z \to X$ to factor through an injection morphism $j : Y \to X$, it is necessary and sufficient that $f(Z) \subset Y$ and, for all $z \in Z$, letting $y = f(z)$, that the homomorphism $(\mathscr{O}_X)_y \to \mathscr{O}_z$ corresponding to $f$ factor as $(\mathscr{O}_X)_y \to (\mathscr{O}_Y)_y \to \mathscr{O}_z$* (or equivalently, *for the kernel of $(\mathscr{O}_X)_y \to \mathscr{O}_z$ to contain the kernel of $(\mathscr{O}_X)_y \to (\mathscr{O}_Y)_y$).*

*Proof.* The conditions are evidently necessary. To see that they are sufficient, we can reduce to the case where $Y$ is a *closed* subprescheme of $X$, by replacing $X$ by an open $U$ such that $Y$ is closed in $U$ (4.1.3) if necessary; $Y$ is then defined by a quasi-coherent sheaf of ideals $\mathscr{I}$ of $\mathscr{O}_X$. Let $f = (\psi, \theta)$, and let $\mathscr{J}$ be the sheaf of ideals of $\psi^*(\mathscr{O}_X)$ which is the kernel of $\theta^\sharp : \psi^*(\mathscr{O}_X) \to \mathscr{O}_Z$; considering the properties of the functor $\psi^*$ (**0**, 3.7.2), the hypotheses imply that, for each $z \in Z$, we have $(\psi^*(\mathscr{I}))_z \subset \mathscr{J}_z$, and, as a result, that $\psi^*(\mathscr{I}) \subset \mathscr{J}$. Thus $\theta^\sharp$ factors as

$$\psi^*(\mathscr{O}_X) \longrightarrow \psi^*(\mathscr{O}_X)/\psi^*(\mathscr{I}) = \psi^*(\mathscr{O}_X/\mathscr{I}) \xrightarrow{\omega} \mathscr{O}_Z,$$

the first arrow being the canonical homomorphism. Let $\psi'$ be the continuous map $Z \to Y$ coinciding with $\psi$; it is clear that we have $\psi'^*(\mathscr{O}_Y) = \psi^*(\mathscr{O}_X/\mathscr{I})$; on the other hand, $\omega$ is evidently a local homomorphism, so $g = (\psi', \omega^\flat)$ is a morphism of preschemes $Z \to Y$ (2.2.1), which, according to the above, is such that $f = j \circ g$, hence the proposition. □ I | 122

**Corollary (4.1.10). —** *For an injection morphism $Z \to X$ to factor through the injection morphism $Y \to X$, it is necessary and sufficient for $Z$ to be a subprescheme of $Y$.*

[8] *[Trans.] There doesn't seem to be an English equivalent of this, except for 'bounded above', which doesn't make much sense in this context. We would normally just say that '$f$ factors through $j$', but to avoid having to entirely restructure the often-lengthy sentences in the original, we sometimes (but as little as we can) use 'majoré'.*

We then write $Z \leqslant Y$, and this condition is evidently an *ordering* on the set of subpreschemes of $X$.

### 4.2. Immersion morphisms

**Definition (4.2.1). —** We say that a morphism $f : Y \to X$ is an immersion (resp. a closed immersion, an open immersion) if it factors as $Y \xrightarrow{g} Z \xrightarrow{j} X$, where $g$ is an isomorphism, $Z$ is a subprescheme of $X$ (resp. a closed subprescheme, a subprescheme induced by an open set), and $j$ is the injection morphism.

The subprescheme $Z$ and the isomorphism $g$ are then determined in a *unique* way, since if $Z'$ is a second subprescheme of $X$, $j'$ the injection $Z' \to X$, and $g'$ an isomorphism $Y \to Z'$ such that $j \circ g = j' \circ g'$, then we have $j' = j \circ g \circ g'^{-1}$, hence $Z' \leqslant Z$ (4.1.10), and we can similarly show that $Z \leqslant Z'$, hence $Z' = Z$, and, since $j$ is a monomorphism of preschemes, $g' = g$.

We say that $f = j \circ g$ is the *canonical factorization* of the immersion $f$, and the subprescheme $Z$ and the isomorphism $g$ are those *associated to* $f$.

It is clear that an immersion is a *monomorphism* of preschemes (4.1.7) and *a fortiori* a *radicial* morphism (3.5.4).

**Proposition (4.2.2). —**

(a) *For a morphism $f = (\psi, \theta) : Y \to X$ to be an open immersion, it is necessary and sufficient for $\psi$ to be a homeomorphism from $Y$ to some open subset of $X$, and, for all $y \in Y$, that the homomorphism $\theta_y^\sharp : \mathscr{O}_{\psi(y)} \to \mathscr{O}_y$ be bijective.*

(b) *For a morphism $f = (\psi, \theta) : Y \to X$ to be an immersion (resp. a closed immersion), it is necessary and sufficient for $\psi$ to be a homeomorphism from $Y$ to some locally closed (resp. closed) subset of $X$, and, for all $y \in Y$, that the homomorphism $\theta_y^\sharp : \mathscr{O}_{\psi(y)} \to \mathscr{O}_y$ be surjective.*

*Proof.*

(a) The conditions are evidently necessary. Conversely, if they are satisfied, then it is clear that $\theta^\sharp$ is an isomorphism from $\psi^*(\mathscr{O}_X)$ to $\mathscr{O}_Y$[9], and $\psi^*(\mathscr{O}_X)$ is the sheaf induced by "transport of structure" via $\psi^{-1}$ from $\mathscr{O}_X|\psi(Y)$; hence the conclusion.

(b) The conditions are evidently necessary—we prove that they are sufficient. Consider first the particular case where we suppose that $X$ is an affine scheme, and that $Z = \psi(Y)$ is *closed* in $X$. We then know (**0**, 3.4.6) that $\psi_*(\mathscr{O}_Y)$ has support equal to $Z$, and that, denoting its restriction to $Z$ by $\mathscr{O}'_Z$, the ringed space $(Z, \mathscr{O}'_Z)$ is induced from $(Y, \mathscr{O}_Y)$ by transport of structure via the homeomorphism $\psi$ considered as a map from $Y$ to $Z$. Let us now show that $f_*(\mathscr{O}_Y) = \psi_*(\mathscr{O}_Y)$ is a *quasi-coherent* $\mathscr{O}_X$-module. Indeed, for all $x \notin Z$, $\psi_*(\mathscr{O}_Y)$ restricted to a suitable neighbourhood of $x$ is zero. If instead $x \in Z$, then we have $x = \psi(y)$ for a unique $y \in Y$; let $V$ be an affine open neighbourhood of $y$ in $Y$; $\psi(V)$ is then open in $Z$, and so equal to the intersection of $Z$ with an open subset $U$ of $X$, and the restriction to $U$ of $\psi_*(\mathscr{O}_Y)$ is identical to the restriction to $U$ of the direct image $(\psi_V)_*(\mathscr{O}_Y|V)$, where $\psi_V$ is the restriction of $\psi$ to $V$. The restriction of the morphism $(\psi, \theta)$ to $(V, \mathscr{O}_Y|V)$ is a morphism from this prescheme to $(X, \mathscr{O}_X)$, and, as a result, is of the form $({}^a\phi, \widetilde{\phi})$, where $\phi$ is the homomorphism from the ring $A = \Gamma(X, \mathscr{O}_X)$ to the ring $\Gamma(V, \mathscr{O}_Y)$ (1.7.3); we conclude that $(\psi_V)_*(\mathscr{O}_Y|V)$ is a quasi-coherent $\mathscr{O}_X$-module (1.6.3), which proves our assertion, due to the local nature of quasi-coherent sheaves. In addition, the hypothesis that $\psi$ is a homeomorphism implies (**0**, 3.4.5) that, for all $y \in Y$, $\psi_y$ is an isomorphism $(\psi_*(\mathscr{O}_Y))_{\psi(y)} \to \mathscr{O}_y$; since the diagram I | 123

$$\begin{tikzcd} \mathscr{O}_{\psi(y)} \arrow[r, "\theta_{\psi(y)}"] \arrow[d, "\psi_y \circ \rho_{\psi(y)}"'] & (\psi_*(\mathscr{O}_Y))_{\psi(y)} \arrow[d, "\psi_y"] \\ (\psi^*(\mathscr{O}_X))_y \arrow[r, "\theta_y^\sharp"] & \mathscr{O}_y \end{tikzcd}$$

[9] *[Trans.] The French source reverses the source and target here. By the definition of a morphism of preschemes and by the displayed stalk maps in Proposition 4.2.2(a), $\theta^\sharp$ has source $\psi^*(\mathscr{O}_X)$ and target $\mathscr{O}_Y$.*

is commutative, and the vertical arrows are the isomorphisms (**0**, 3.7.2), the hypothesis that $\theta_y^\sharp$ is surjective implies that $\theta_{\psi(y)}$ is surjective as well. Since the support of $\psi_*(\mathscr{O}_Y)$ is $Z = \psi(Y)$, $\theta$ is a *surjective* homomorphism from $\mathscr{O}_X = \widetilde{A}$ to the quasi-coherent $\mathscr{O}_X$-module $f_*(\mathscr{O}_Y)$. As a result, there exists a unique isomorphism $\omega$ from a sheaf quotient $\widetilde{A}/\widetilde{\mathfrak{J}}$ (with $\mathfrak{J}$ being an ideal of $A$) to $f_*(\mathscr{O}_Y)$ which, when composed with the canonical homomorphism $\widetilde{A} \to \widetilde{A}/\widetilde{\mathfrak{J}}$, gives $\theta$ (1.3.8); if $\mathscr{O}_Z$ denotes the restriction of $\widetilde{A}/\widetilde{\mathfrak{J}}$ to $Z$, then $(Z, \mathscr{O}_Z)$ is a subprescheme of $(X, \mathscr{O}_X)$, and $f$ factors through the canonical injection of this subprescheme into $X$ and the isomorphism $(\psi_0, \omega_0)$, where $\psi_0$ is $\psi$ considered as a map from $Y$ to $Z$, and $\omega_0$ the restriction of $\omega$ to $\mathscr{O}_Z$.

We now pass to the general case. Let $U$ be an affine open subset of $X$ such that $U \cap \psi(Y)$ is closed in $U$ and nonempty. By restricting $f$ to the prescheme induced by $Y$ on the open subset $\psi^{-1}(U)$, and by considering it as a morphism from this prescheme to the prescheme induced by $X$ on $U$, we reduce to the first case; the restriction of $f$ to $\psi^{-1}(U)$ is thus a closed immersion $\psi^{-1}(U) \to U$, canonically factoring as $j_U \circ g_U$, where $g_U$ is an isomorphism from the prescheme $\psi^{-1}(U)$ to a subprescheme $Z_U$ of $U$, and $j_U$ is the canonical injection $Z_U \to U$. Let $V$ be a second affine open subset of $X$ such that $V \subset U$; since the restriction $Z'_V$ of $Z_U$ to $V$ is a subprescheme of the prescheme $V$, the restriction of $f$ to $\psi^{-1}(V)$ factors as $j'_V \circ g'_V$, where $j'_V$ is the canonical injection $Z'_V \to V$ and $g'_V$ is an isomorphism from $\psi^{-1}(V)$ to $Z'_V$. By the uniqueness of the canonical factorization of an immersion (4.2.1), we necessarily have that $Z'_V = Z_V$ and $g'_V = g_V$. We conclude (4.1.5) that there is a subprescheme $Z$ of $X$ whose underlying space is $\psi(Y)$, and whose restriction to each $U \cap \psi(Y)$ is $Z_U$; the $g_U$ are then the restrictions to $\psi^{-1}(U)$ of an isomorphism $g : Y \to Z$ such that $f = j \circ g$, where $j$ is the canonical injection $Z \to X$.

□

**Corollary (4.2.3). —** *Let $X$ be an affine scheme. For a morphism $f = (\psi, \theta) : Y \to X$ to be a closed immersion, it is necessary and sufficient for $Y$ to be an affine scheme, and the homomorphism $\Gamma(\psi) : \Gamma(\mathscr{O}_X) \to \Gamma(\mathscr{O}_Y)$ to be surjective.*

**Corollary (4.2.4). —**

(a) *Let $f$ be a morphism $Y \to X$, and $(V_\lambda)$ a cover of $f(Y)$ by open subsets of $X$. For $f$ to be an immersion (resp. an open immersion), it is necessary and sufficient for its restriction to each of the induced preschemes $f^{-1}(V_\lambda)$ to be an immersion (resp. an open immersion) into $V_\lambda$.* I | 124

(b) *Let $f$ be a morphism $Y \to X$, and $(V_\lambda)$ an open cover of $X$. For $f$ to be a closed immersion, it is necessary and sufficient for its restriction to each of the induced preschemes $f^{-1}(V_\lambda)$ to be a closed immersion into $V_\lambda$.*

*Proof.* Let $f = (\psi, \theta)$; in the case (a), $\theta_y^\sharp$ is surjective (resp. bijective) for all $y \in Y$, and in the case (b), $\theta_y^\sharp$ is surjective for all $y \in Y$; it thus suffices to check, in case (a), that $\psi$ is a homeomorphism from $Y$ to a locally closed (resp. open) subset of $X$, and, in case (b), that $\psi$ is a homeomorphism from $Y$ to a closed subset of $X$. Now $\psi$ is evidently injective, and sends each neighbourhood of $y$ in $Y$ to a neighbourhood of $\psi(y)$ in $\psi(Y)$ for all $y \in Y$, by virtue of the hypothesis; in case (a), $\psi(Y) \cap V_\lambda$ is locally closed (resp. open) in $V_\lambda$, so $\psi(Y)$ is locally closed (resp. open) in the union of the $V_\lambda$, and *a fortiori* in $X$; in case (b), $\psi(Y) \cap V_\lambda$ is closed in $V_\lambda$, so $\psi(Y)$ is closed in $X$ since $X = \bigcup_\lambda V_\lambda$. □

**Proposition (4.2.5). —** *The composition of any two immersions (resp. of two open immersions, of two closed immersions) is an immersion (resp. an open immersion, a closed immersion).*

*Proof.* This follows easily from (4.1.6). □

### 4.3. Products of immersions

**Proposition (4.3.1).** — *Let $\alpha : X' \to X$, $\beta : Y' \to Y$ be two S-morphisms; if $\alpha$ and $\beta$ are immersions (resp. open immersions, closed immersions), then $\alpha \times_S \beta$ is an immersion (resp. an open immersion, a closed immersion). In addition, if $\alpha$ (resp. $\beta$) identifies $X'$ (resp. $Y'$) with a subprescheme $X''$ (resp. $Y''$) of $X$ (resp. $Y$), then $\alpha \times_S \beta$ identifies the underlying space of $X' \times_S Y'$ with the subspace $p^{-1}(X'') \cap q^{-1}(Y'')$ of the underlying space of $X \times_S Y$, where $p$ and $q$ denote the projections from $X \times_S Y$ to $X$ and $Y$ respectively.*

*Proof.* According to Definition (4.2.1), we can restrict to the case where $X'$ and $Y'$ are subpreschemes, and $\alpha$ and $\beta$ are the injection morphisms. The proposition has already been proved for the subpreschemes induced by open sets (3.2.7); since every subprescheme is a closed subprescheme of a prescheme induced by an open set (4.1.3), we can reduce to the case where $X'$ and $Y'$ are *closed* subpreschemes.

Let us first show that we can assume $S$ is *affine*. Let $(S_\lambda)$ be a cover of $S$ by affine open sets; if $\phi$ and $\psi$ are the structure morphisms of $X$ and $Y$, then let $X_\lambda = \phi^{-1}(S_\lambda)$ and $Y_\lambda = \psi^{-1}(S_\lambda)$. The restriction $X'_\lambda$ (resp. $Y'_\lambda$) of $X'$ (resp. $Y'$) to $X_\lambda \cap X'$ (resp. $Y_\lambda \cap Y'$) is a closed subprescheme of $X_\lambda$ (resp. $Y_\lambda$), the preschemes $X_\lambda$, $Y_\lambda$, $X'_\lambda$, and $Y'_\lambda$ can be considered as $S_\lambda$-preschemes, and the products $X_\lambda \times_S Y_\lambda$ and $X_\lambda \times_{S_\lambda} Y_\lambda$ (resp. $X'_\lambda \times_S Y'_\lambda$ and $X'_\lambda \times_{S_\lambda} Y'_\lambda$) are identical (3.2.5). If the proposition is true when $S$ is affine, then the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\lambda$ is thus an immersion (3.2.7). Since the product $X'_\lambda \times_S Y'_\mu$ (resp. $X_\lambda \times_S Y_\mu$) can be identified with $(X'_\lambda \cap X'_\mu) \times_S (Y'_\lambda \cap Y'_\mu)$ (resp. $(X_\lambda \cap X_\mu) \times_S (Y_\lambda \cap Y_\mu)$) (3.2.6.4), the restriction of $\alpha \times_S \beta$ to each of the $X'_\lambda \times_S Y'_\mu$ is also an I | 125
immersion; the same is true for $\alpha \times_S \beta$ by (4.2.4).

Next, we show that we can assume $X$ and $Y$ are *affine*. Indeed, let $(U_i)$ (resp. $(V_j)$) be a cover of $X$ (resp. $Y$) by affine open sets, and let $X'_i$ (resp. $Y'_j$) be the restriction of $X'$ (resp. $Y'$) to $X' \cap U_i$ (resp. $Y' \cap V_j$), which is a closed subprescheme of $U_i$ (resp. $V_j$); $U_i \times_S V_j$ can be identified with the restriction of $X \times_S Y$ to $p^{-1}(U_i) \cap q^{-1}(V_j)$ (3.2.7); similarly, if $p'$ and $q'$ are the projections from $X' \times_S Y'$, then $X'_i \times_S Y'_j$ can be identified with the restriction of $X' \times_S Y'$ to ${p'}^{-1}(X'_i) \cap {q'}^{-1}(Y'_j)$. Set $\gamma = \alpha \times_S \beta$; we have, by definition, $p \circ \gamma = \alpha \circ p'$ and $q \circ \gamma = \beta \circ q'$; since $X'_i = \alpha^{-1}(U_i)$ and $Y'_j = \beta^{-1}(V_j)$, we also have ${p'}^{-1}(X'_i) = \gamma^{-1}(p^{-1}(U_i))$ and ${q'}^{-1}(Y'_j) = \gamma^{-1}(q^{-1}(V_j))$, hence

$$ {p'}^{-1}(X'_i) \cap {q'}^{-1}(Y'_j) = \gamma^{-1}(p^{-1}(U_i) \cap q^{-1}(V_j)) = \gamma^{-1}(U_i \times_S V_j), $$

and we then conclude as in the previous part of the proof.

So suppose $X$, $Y$, and $S$ are affine, and let $B$, $C$, and $A$ be their respective rings. Then $B$ and $C$ are $A$-algebras, and $X'$ and $Y'$ are affine schemes whose rings are quotient algebras $B'$ and $C'$ of $B$ and $C$ respectively. In addition, we have $\alpha = ({}^a\rho, \widetilde{\rho})$ and $\beta = ({}^a\sigma, \widetilde{\sigma})$, where $\rho$ and $\sigma$ are (respectively) the canonical homomorphisms $B \to B'$ and $C \to C'$ (1.7.3). With that in mind, we know that $X \times_S Y$ (resp. $X' \times_S Y'$) is an affine scheme with ring $B \otimes_A C$ (resp. $B' \otimes_A C'$), and $\alpha \times_S \beta = ({}^a\tau, \widetilde{\tau})$, where $\tau$ is the homomorphism $\rho \otimes \sigma$ from $B \otimes_A C$ to $B' \otimes_A C'$ (Proposition (3.2.2) and Corollary (3.2.3)); since this homomorphism is surjective, $\alpha \times_S \beta$ is an immersion. In addition, if $\mathfrak{b}$ (resp. $\mathfrak{c}$) is the kernel of $\rho$ (resp. $\sigma$), then the kernel of $\tau$ is $u(\mathfrak{b}) + v(\mathfrak{c})$, where $u$ (resp. $v$) is the homomorphism $b \mapsto b \otimes 1$ (resp. $c \mapsto 1 \otimes c$). Since $p = ({}^a u, \widetilde{u})$ and $q = ({}^a v, \widetilde{v})$, this kernel corresponds, in the prime spectrum of $B \otimes_A C$, to the closed set $p^{-1}(X') \cap q^{-1}(Y')$, by (1.2.2.1) and Proposition (1.1.2, iii), which finishes the proof. □

**Corollary (4.3.2).** — *If $f : X \to Y$ is an immersion (resp. an open immersion, a closed immersion) and an S-morphism, then $f_{(S')}$ is an immersion (resp. an open immersion, a closed immersion) for every extension $S' \to S$ of the base prescheme.*

### 4.4. Inverse images of a subprescheme

**Proposition (4.4.1).** — *Let $f : X \to Y$ be a morphism, $Y'$ a subprescheme (resp. a closed subprescheme, a prescheme induced by an open set) of $Y$, and $j : Y' \to Y$ the injection morphism. Then the projection $p : X \times_Y Y' \to X$ is an immersion (resp. a closed immersion, an open immersion); the underlying space of the subprescheme of $X$ associated to $p$ is $f^{-1}(Y')$; in addition, if $j'$ is the injection morphism of this subprescheme into $X$, then, for every morphism $h : Z \to X$, the composite $f \circ h : Z \to Y$ factors through $j$ if and only if $h$ factors through $j'$.*

*Proof.* Since $p = 1_X \times_Y j$ (3.3.4), the first claim follows from Proposition (4.3.1); the second is a particular case of Corollary (3.5.10) (after swapping the roles of $X$ and $Y'$). Finally, if we have $f \circ h = j \circ h'$, where $h'$ is a morphism $Z \to Y'$, then it follows from the definition of the product that we have $h = p \circ u$, where $u$ is a morphism $Z \to X \times_Y Y'$, whence the last claim. □

We say that the subprescheme of $X$ thus defined is the *inverse image* of the subprescheme $Y'$ of $Y$ under the morphism $f$, terminology which is consistent with that introduced more generally in (3.3.6). When we speak of $f^{-1}(Y')$ as a subprescheme of $X$, this will always be the subprescheme we mean. **I** | 126

When the preschemes $f^{-1}(Y')$ and $X$ are identical, $j'$ is the identity and each morphism $h : Z \to X$ thus factors through $j'$; taking $h = 1_X$, we see that *the morphism $f : X \to Y$ factors as $X \xrightarrow{g} Y' \xrightarrow{j} Y$.*

When $y$ is a *closed* point of $Y$ and $Y' = \mathrm{Spec}(k(y))$ is the smallest closed subprescheme of $Y$ having $\{y\}$ as its underlying space (4.1.9), the closed subprescheme $f^{-1}(Y')$ is canonically isomorphic to the *fibre* $f^{-1}(y)$ defined in (3.6.2), and we will use this identification in all that follows.

**Corollary (4.4.2).** — *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms, and $h = g \circ f$ their composition. For each subprescheme $Z'$ of $Z$, the subpreschemes $f^{-1}(g^{-1}(Z'))$ and $h^{-1}(Z')$ of $X$ are identical.*

*Proof.* This follows from the existence of the canonical isomorphism $X \times_Y (Y \times_Z Z') \simeq X \times_Z Z'$ (3.3.9.1). □

**Corollary (4.4.3).** — *Let $X'$ and $X''$ be subpreschemes of $X$, and let $j' : X' \to X$ and $j'' : X'' \to X$ be their injection morphisms; then ${j'}^{-1}(X'')$ and ${j''}^{-1}(X')$ are both equal to the greatest lower bound* $\inf(X', X'')$ *of $X'$ and $X''$ for the ordering $\leqslant$ on subpreschemes, and this is canonically isomorphic to $X' \times_X X''$.*

*Proof.* This follows immediately from Proposition (4.4.1) and Corollary (4.1.10). □

**Corollary (4.4.4).** — *Let $f : X \to Y$ be a morphism, and $Y'$ and $Y''$ subpreschemes of $Y$; then we have $f^{-1}(\inf(Y', Y'')) = \inf(f^{-1}(Y'), f^{-1}(Y''))$.*

*Proof.* This follows from the existence of the canonical isomorphism between $(X \times_Y Y') \times_X (X \times_Y Y'')$ and $X \times_Y (Y' \times_Y Y'')$ (3.3.9.1). □

**Proposition (4.4.5).** — *Let $f : X \to Y$ be a morphism, and $Y'$ a closed subprescheme of $Y$ defined by a quasi-coherent sheaf of ideals $\mathscr{K}$ of $\mathscr{O}_Y$ (4.1.3); the closed subprescheme $f^{-1}(Y')$ of $X$ is then defined by the quasi-coherent sheaf of ideals $f^*(\mathscr{K})\mathscr{O}_X$ of $\mathscr{O}_X$.*

*Proof.* The statement is evidently local on $X$ and $Y$; it thus suffices to note that if $A$ is a $B$-algebra and $\mathfrak{K}$ an ideal of $B$, then we have $A \otimes_B (B/\mathfrak{K}) = A/\mathfrak{K}A$, and then to apply (1.6.9). □ **Err$_{\mathrm{II}}$**

**Corollary (4.4.6).** — *Let $X'$ be a closed subprescheme of $X$ defined by a quasi-coherent sheaf of ideals $\mathscr{J}$ of $\mathscr{O}_X$, and $i$ the injection $X' \to X$; for the restriction $f \circ i$ of $f$ to $X'$ to factor through the injection $j : Y' \to Y$* (in other words, for it to factor as $j \circ g$, with $g$ a morphism $X' \to Y'$), *it is necessary and sufficient that $f^*(\mathscr{K})\mathscr{O}_X \subset \mathscr{J}$.*

*Proof.* It suffices to apply Proposition (4.4.1) to $i$, taking Proposition (4.4.5) into account. □

### 4.5. Local immersions and local isomorphisms

**Definition (4.5.1). —** Let $f : X \to Y$ be a morphism of preschemes. We say that $f$ is a local immersion at a point $x \in X$ if there exists an open neighbourhood $U$ of $x$ in $X$ and an open neighbourhood $V$ of $f(x)$ in $Y$ such that the restriction of $f$ to the induced prescheme $U$ is a closed immersion of $U$ into the induced prescheme $V$. We say that $f$ is a local immersion if $f$ is a local immersion at each point of $X$.

**Definition (4.5.2). —** We say that a morphism $f : X \to Y$ is a local isomorphism at a point $x \in X$ if there exists an open neighbourhood $U$ of $x$ in $X$ such that the restriction of $f$ to the induced prescheme $U$ is an open immersion of $U$ into $Y$. We say that $f$ is a local isomorphism if $f$ is a local isomorphism at each point of $X$. **I** | 127

**(4.5.3).** An immersion (resp. a closed immersion) $f : X \to Y$ can be characterized as a local immersion such that $f$ is a homeomorphism from the underlying space of $X$ to a subset (resp. a closed subset) of $Y$. An open immersion $f$ can be characterized as an *injective* local isomorphism.

**Proposition (4.5.4). —** *Let $X$ be an irreducible prescheme, and $f : X \to Y$ a dominant injective morphism. If $f$ is a local immersion, then $f$ is an immersion, and $f(X)$ is open in $Y$.*

*Proof.* Let $x \in X$, and choose an open neighbourhood $U$ of $x$ and an open neighbourhood $V$ of $f(x)$ in $Y$ such that the restriction of $f$ to $U$ is a closed immersion into $V$; since $U$ is dense in $X$, $f(U)$ is dense in $Y$ by hypothesis, so $f(U) = V$, and $f$ is a homeomorphism from $U$ to $V$; the hypothesis that $f$ is injective implies that $f^{-1}(V) = U$, hence the proposition. □

**Proposition (4.5.5). —**

(i) *The composition of any two local immersions (resp. of two local isomorphisms) is a local immersion (resp. a local isomorphism).*

(ii) *Let $f : X \to X'$ and $g : Y \to Y'$ be two $S$-morphisms. If $f$ and $g$ are local immersions (resp. local isomorphisms), then so too is $f \times_S g$.*

(iii) *If an $S$-morphism $f$ is a local immersion (resp. a local isomorphism), then so too is $f_{(S')}$ for every extension $S' \to S$ of the base prescheme.*

*Proof.* According to (3.5.1), it suffices to prove (i) and (ii).

(i) follows immediately from the transitivity of closed (resp. open) immersions (4.2.5)[10] and from the fact that if $f$ is a homeomorphism from $X$ to a closed subset of $Y$, then for every open $U \subset X$, $f(U)$ is open in $f(X)$, so there exists an open subset $V$ of $Y$ such that $f(U) = V \cap f(X)$, and, as a result, $f(U)$ is closed in $V$.

To prove (ii), let $z \in X \times_S Y$, set $z' = (f \times_S g)(z)$[11], let $p$ and $q$ be the projections from $X \times_S Y$, and let $p'$ and $q'$ be the projections from $X' \times_S Y'$. There exist, by hypothesis, open neighbourhoods $U$, $U'$, $V$, and $V'$ of $x = p(z)$, $x' = p'(z')$, $y = q(z)$, and $y' = q'(z')$ (respectively), such that the restrictions of $f$ and $g$ to $U$ and $V$ (respectively) are closed (resp. open) immersions into $U'$ and $V'$ (respectively). Since the underlying spaces of $U \times_S V$ and $U' \times_S V'$ can be identified with the open neighbourhoods $p^{-1}(U) \cap q^{-1}(V)$ and $p'^{-1}(U') \cap q'^{-1}(V')$ of $z$ and $z'$ (respectively) (3.2.7), the proposition follows from Proposition (4.3.1). □

## §5. Reduced preschemes; the separation condition

### 5.1. Reduced preschemes

**Proposition (5.1.1). —** *Let $(X, \mathscr{O}_X)$ be a prescheme, and $\mathscr{B}$ a quasi-coherent $\mathscr{O}_X$-algebra. Then there exists a unique quasi-coherent $\mathscr{O}_X$-module $\mathscr{N}$ whose stalk $\mathscr{N}_x$ at any $x \in X$ is the nilradical of the ring $\mathscr{B}_x$. When $X$ is affine, and, consequently, $\mathscr{B} = \widetilde{B}$, where $B$ is an algebra over $A(X)$, then we have $\mathscr{N} = \widetilde{\mathfrak{N}}$, where $\mathfrak{N}$ is the nilradical of $B$.*

[10] *[Trans.] The French source cites (4.2.4) here; transitivity is Proposition 4.2.5.*

[11] *[Trans.] The French source uses $z$ and $z'$ below without introducing them; this insertion supplies the required point and its image.*

*Proof.* The statement is local, so it suffices to show the latter claim. We know that $\widetilde{\mathfrak{N}}$ is a quasi-coherent $\mathscr{O}_X$-module (1.4.1), and that its stalk at a point $x \in X$ is the ideal $\mathfrak{N}_x$ of the ring of fractions $B_x$; it remains to prove that the nilradical of $B_x$ is contained in $\mathfrak{N}_x$, the converse inclusion being evident. Let $z/s$ be an element of the nilradical of $B_x$, with $z \in B$, and $s \notin \mathfrak{j}_x$; by hypothesis, there exists an integer $k$ such that $(z/s)^k = 0$, which implies that there exists some $t \notin \mathfrak{j}_x$ such that $tz^k = 0$. We conclude that $(tz)^k = 0$, and, as a result, that $z/s = (tz)/(ts) \in \mathfrak{N}_x$. ☐ **I** | 128

We say that the quasi-coherent $\mathscr{O}_X$-module $\mathscr{N}$ thus defined is the *nilradical* of the $\mathscr{O}_X$-algebra $\mathscr{B}$; in particular, we denote by $\mathscr{N}_X$ the nilradical of $\mathscr{O}_X$.

**Corollary (5.1.2). —** *Let $X$ be a prescheme; the closed subprescheme of $X$ defined by the sheaf of ideals $\mathscr{N}_X$ is the only reduced subprescheme* (**0**, 4.1.4) *of $X$ that has $X$ as its underlying space; it is also the smallest subprescheme of $X$ that has $X$ as its underlying space.*

*Proof.* Let $Y$ denote the closed subprescheme of $X$ defined by $\mathscr{N}_X$. Since its structure sheaf is $\mathscr{O}_X/\mathscr{N}_X$, it is immediate that $Y$ is reduced and has $X$ as its underlying space, because $\mathscr{N}_x \neq \mathscr{O}_x$ for every $x \in X$. To show the other claims, note that a subprescheme $Z$ of $X$ that has $X$ as its underlying space is defined by a sheaf of ideals $\mathscr{I}$ (4.1.3) such that $\mathscr{I}_x \neq \mathscr{O}_x$ for any $x \in X$. We can restrict to the case where $X$ is affine, say $X = \operatorname{Spec}(A)$ and $\mathscr{I} = \widetilde{\mathfrak{I}}$, where $\mathfrak{I}$ is an ideal of $A$; then, for every $x \in X$, we have $\mathfrak{I} \subset \mathfrak{j}_x$, and so $\mathfrak{I}$ is contained in every prime ideal of $A$, and so also in their intersection $\mathfrak{N}$, the nilradical of $A$. This proves that $Y$ is the smallest subprescheme of $X$ that has $X$ as its underlying space (4.1.9); furthermore, if $Z$ is distinct from $Y$, we necessarily have $\mathscr{I}_x \neq \mathscr{N}_x$ for at least one $x \in X$, and so (5.1.1) $Z$ is not reduced. ☐

**Definition (5.1.3). —** We define the reduced prescheme associated to a prescheme $X$, denoted by $X_{\mathrm{red}}$, to be the unique reduced subprescheme of $X$ that has $X$ as its underlying space.

Saying that a prescheme $X$ is reduced thus means that $X = X_{\mathrm{red}}$.

**Proposition (5.1.4). —** *For the prime spectrum of a ring $A$ to be a reduced (resp. integral) prescheme* (2.1.7)*, it is necessary and sufficient for $A$ to be a reduced (resp. integral) ring.*

*Proof.* Indeed, it follows immediately from (5.1.1) that the condition $\mathscr{N} = (0)$ is necessary and sufficient for $X = \operatorname{Spec}(A)$ to be reduced; the claim corresponding to integral rings is then a consequence of (1.1.13). ☐

Since every nonzero ring of fractions of an integral ring is integral, it follows from (5.1.4) that, for every *locally integral* prescheme $X$, $\mathscr{O}_x$ is an *integral* ring for every $x \in X$. The converse is true whenever the underlying space of $X$ is *locally Noetherian*: indeed, $X$ is then reduced, and if $U$ is an affine open subset of $X$, which is a Noetherian space, then $U$ has only a finite number of irreducible components, and so its ring $A$ has only a finite number of minimal prime ideals (1.1.14). If two of the irreducible components of $U$ had a common point $x$, then $\mathscr{O}_x$ would have at least two distinct minimal prime ideals, and would thus not be integral; the components of $U$ are thus open subsets that are pairwise disjoint, and each of them is thus integral.

**(5.1.5).** Let $f = (\psi, \theta) : X \to Y$ be a morphism of preschemes; the homomorphism $\theta_x^\sharp : \mathscr{O}_{\psi(x)} \to \mathscr{O}_x$ sends each nilpotent element of $\mathscr{O}_{\psi(x)}$ to a nilpotent element of $\mathscr{O}_x$; by passing to the quotients, $\theta^\sharp$ induces a homomorphism **I** | 129

$$\omega : \psi^*(\mathscr{O}_Y/\mathscr{N}_Y) \longrightarrow \mathscr{O}_X/\mathscr{N}_X.$$

It is clear that, for every $x \in X$, $\omega_x : \mathscr{O}_{\psi(x)}/\mathscr{N}_{\psi(x)} \to \mathscr{O}_x/\mathscr{N}_x$ is a local homomorphism, and so $(\psi, \omega^\flat)$ is a morphism of preschemes $X_{\mathrm{red}} \to Y_{\mathrm{red}}$, which we denote by $f_{\mathrm{red}}$, and call the *reduced* morphism associated to $f$. It is immediate that, for morphisms $f : X \to Y$ and $g : Y \to Z$, we have $(g \circ f)_{\mathrm{red}} = g_{\mathrm{red}} \circ f_{\mathrm{red}}$, and so we have defined $X_{\mathrm{red}}$ as a *functor*, *covariant* in $X$.

The preceding definition shows that the diagram

$$\begin{tikzcd} X_{\mathrm{red}} \arrow[r, "f_{\mathrm{red}}"] \arrow[d] & Y_{\mathrm{red}} \arrow[d] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

is commutative, where the vertical arrows are the injection morphisms; in other words, $X_{\text{red}} \to X$ is a *functorial* morphism. We note in particular that, if $X$ is reduced, then every morphism $f : X \to Y$ factors as $X \xrightarrow{f_{\text{red}}} Y_{\text{red}} \to Y$; in other words, $f$ factors through the injection morphism $Y_{\text{red}} \to Y$.

**Proposition (5.1.6).** — *Let $f : X \to Y$ be a morphism; if $f$ is surjective (resp. radicial, an immersion, a closed immersion, an open immersion, a local immersion, a local isomorphism), then so too is $f_{\text{red}}$. Conversely, if $f_{\text{red}}$ is surjective (resp. radicial), then so too is $f$.*

*Proof.* The proposition is trivial if $f$ is surjective; if $f$ is radicial, then the proposition follows from the fact that, for every $x \in X$, the field $k(x)$ is the same for the preschemes $X$ and $X_{\text{red}}$ (3.5.8). Finally, if $f = (\psi, \theta)$ is an immersion, a closed immersion, or a local immersion (resp. an open immersion, or a local isomorphism), then the proposition follows from the fact that, if $\theta_x^\sharp$ is surjective (resp. bijective), then so too is the homomorphism obtained by passing to the quotients by the nilradicals $\mathscr{O}_{\psi(x)}$ and $\mathscr{O}_x$ ((5.1.2) and (4.2.2)) (cf. (5.5.12)). □

**Proposition (5.1.7).** — *If $X$ and $Y$ are $S$-preschemes, then the preschemes $X_{\text{red}} \times_{S_{\text{red}}} Y_{\text{red}}$ and $X_{\text{red}} \times_S Y_{\text{red}}$ are identical, and canonically identified with a subprescheme of $X \times_S Y$ that has the same underlying space as that product.*

*Proof.* The canonical identification of $X_{\text{red}} \times_S Y_{\text{red}}$ with a subprescheme of $X \times_S Y$ that has the same underlying space follows from (4.3.1). Furthermore, if $\phi$ and $\psi$ are the structure morphisms $X_{\text{red}} \to S$ and $Y_{\text{red}} \to S$ (respectively), then they factor through $S_{\text{red}}$ (5.1.5), and since $S_{\text{red}} \to S$ is a monomorphism, the first claim of the proposition follows from (3.2.4). □

**Corollary (5.1.8).** — *The preschemes $(X \times_S Y)_{\text{red}}$ and $(X_{\text{red}} \times_{S_{\text{red}}} Y_{\text{red}})_{\text{red}}$ are canonically identified with one another.*

*Proof.* This follows from (5.1.2) and (5.1.7). □

We note that, even if $X$ and $Y$ are reduced preschemes, $X \times_S Y$ might not be reduced, because the tensor product of two reduced algebras can have nilpotent elements.

**Proposition (5.1.9).** — *Let $X$ be a prescheme, and $\mathscr{I}$ a quasi-coherent sheaf of ideals of $\mathscr{O}_X$ such that $\mathscr{I}^n = 0$ for some integer $n > 0$. Let $X_0$ be the closed subprescheme $(X, \mathscr{O}_X/\mathscr{I})$ of $X$; for $X$ to be an affine scheme, it is necessary and sufficient for $X_0$ to be an affine scheme.* I | 130

The condition is clearly necessary, so we will show that it is sufficient. If we set $X_k = (X, \mathscr{O}_X/\mathscr{I}^{k+1})$, it is enough to prove by induction on $k$ that $X_k$ is affine, and so we are led to consider the base case, where $\mathscr{I}^2 = 0$. We set

$$\begin{aligned} A &= \Gamma(X, \mathscr{O}_X) \\ A_0 &= \Gamma(X_0, \mathscr{O}_{X_0}) = \Gamma(X, \mathscr{O}_X/\mathscr{I}). \end{aligned}$$

The canonical homomorphism $\mathscr{O}_X \to \mathscr{O}_X/\mathscr{I}$ induces a homomorphism of rings $\phi : A \to A_0$. We will see below that $\phi$ is *surjective*, which implies that

(5.1.9.1) $$0 \longrightarrow \Gamma(X, \mathscr{I}) \longrightarrow \Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(X, \mathscr{O}_X/\mathscr{I}) \longrightarrow 0$$

is an *exact* sequence. We now prove, assuming that this is true, the proposition. Note that $\mathfrak{K} = \Gamma(X, \mathscr{I})$ is an ideal whose square is zero in $A$, and thus a module over $A_0 = A/\mathfrak{K}$. By hypothesis, we have $X_0 = \operatorname{Spec}(A_0)$, and, since the underlying spaces of $X_0$ and $X$ are identical, $\mathfrak{K} = \Gamma(X_0, \mathscr{I})$; Additionally, since $\mathscr{I}^2 = 0$, $\mathscr{I}$ is a quasi-coherent $(\mathscr{O}_X/\mathscr{I})$-module, so we have $\mathscr{I} \cong \widetilde{\mathfrak{K}}$ and $\mathfrak{K}_x = \mathscr{I}_x$ for all $x \in X_0$ (1.4.1). With this in mind, let $X' = \operatorname{Spec}(A)$, and consider the morphism $f = (\psi, \theta) : X \to X'$ of preschemes that corresponds to the identity map $A \to \Gamma(X, \mathscr{O}_X)$ (2.2.4). For every affine open subset $V$ of $X$, the diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & \Gamma(V, \mathscr{O}_X|V) \arrow[d] \\ A_0 = A/\mathfrak{K} \arrow[r] & \Gamma(V, \mathscr{O}_{X_0}|V) \end{tikzcd}$$

commutes, whence the diagram

$$\begin{tikzcd} X' & X \arrow[l, "f"'] \\ X_0' \arrow[u, "j'"] & X_0 \arrow[l, "f_0"] \arrow[u, "j"'] \end{tikzcd}$$

also commutes ($X_0'$ being the closed subprescheme of $X'$ defined by the quasi-coherent sheaf of ideals $\widetilde{\mathfrak{K}}$, and $j$ and $j'$ being the canonical injection morphisms). But since $X_0$ is affine, $f_0$ is an isomorphism, and since the underlying continuous maps of $j$ and $j'$ are identity maps, we see straight away that $\psi : X \to X'$ is a homeomorphism. Furthermore, the equation $\mathfrak{K}_x = \mathscr{I}_x$ shows that the restriction of $\theta^\sharp : \psi^*(\mathscr{O}_{X'}) \to \mathscr{O}_X$ is an *isomorphism* from $\psi^*(\widetilde{\mathfrak{K}})$ to $\mathscr{I}$; additionally, by passing to the quotients, $\theta^\sharp$ gives an *isomorphism* $\psi^*(\mathscr{O}_{X'}/\widetilde{\mathfrak{K}}) \to \mathscr{O}_X/\mathscr{I}$, because $f_0$ is an isomorphism; we thus immediately conclude, by the 5 lemma (M, I, 1.1), that $\theta^\sharp$ is itself an isomorphism, and thus that $f$ is an *isomorphism*, and thus that $X$ is affine. So everything reduces to proving the exactness of (5.1.9.1), which will follow from showing that $\mathrm{H}^1(X, \mathscr{I}) = 0$. But $\mathrm{H}^1(X, \mathscr{I}) = \mathrm{H}^1(X_0, \mathscr{I})$, and we **I** | 131
have seen that $\mathscr{I}$ is a quasi-coherent $\mathscr{O}_{X_0}$-module. Our proof will thus follow from

**Lemma (5.1.9.2). —** *If $Y$ is an affine scheme, and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_Y$-module, then $\mathrm{H}^1(Y, \mathscr{F}) = 0$.*

*Proof.* This lemma will be proven in Chapter III, ğ1, as a consequence of the more general theorem that $\mathrm{H}^i(Y, \mathscr{F}) = 0$ for all $i > 0$. To give an independent proof, note that $\mathrm{H}^1(Y, \mathscr{F})$ can be identified with the module $\mathrm{Ext}^1_{\mathscr{O}_Y}(Y; \mathscr{O}_Y, \mathscr{F})$ of extension classes of the $\mathscr{O}_Y$-module $\mathscr{O}_Y$ by the $\mathscr{O}_Y$-module $\mathscr{F}$ (T, 4.2.3); so everything reduces to proving that such an extension $\mathscr{G}$ is trivial. But, for every $y \in Y$, there is a neighbourhood $V$ of $y$ in $Y$ such that $\mathscr{G}|V$ is isomorphic to $\mathscr{F}|V \oplus \mathscr{O}_Y|V$[12] (**0**, 5.4.9); from this we conclude that $\mathscr{G}$ is a *quasi-coherent* $\mathscr{O}_Y$-module. If $A$ is the ring of $Y$, then we have $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$, where $M$ and $N$ are $A$-modules, and, by hypothesis, $N$ is an extension of the $A$-module $A$ by the $A$-module $M$ (1.3.11). Since this extension is necessarily trivial, the lemma is proven, and thus so too is (5.1.9). □

**Corollary (5.1.10). —** *Let $X$ be a prescheme such that $\mathscr{N}_X$ is nilpotent. For $X$ to be an affine scheme, it is necessary and sufficient for $X_{\mathrm{red}}$ to be an affine scheme.*

### 5.2. Existence of a subprescheme with a given underlying space

**Proposition (5.2.1). —** *For every locally closed subspace $Y$ of the underlying space of a prescheme $X$, there exists exactly one reduced subprescheme of $X$ that has $Y$ as its underlying space.*

*Proof.* The uniqueness follows from (5.1.2), so it remains only to show the existence of the prescheme in question.

If $X$ is affine with ring $A$, and $Y$ is closed in $X$, then the proposition is immediate: $\mathfrak{j}(Y)$ is the largest ideal $\mathfrak{a} \subset A$ such that $V(\mathfrak{a}) = Y$, and it is equal to its radical (1.1.4, i), so $A/\mathfrak{j}(Y)$ is a reduced ring.

In the general case, for every affine open $U \subset X$ such that $U \cap Y$ is closed in $U$, consider the closed subprescheme $Y_U$ of $U$ defined by the sheaf of ideals associated to the ideal $\mathfrak{j}(U \cap Y)$ of $A(U)$. This subprescheme is reduced. We claim that, if $V$ is an affine open subset of $X$ contained in $U$, then $Y_V$ is *induced* by $Y_U$ on $V \cap Y$. Indeed, this induced prescheme is a closed reduced subprescheme of $V$ having $V \cap Y$ as its underlying space; the uniqueness of $Y_V$ therefore proves the claim. □

**Proposition (5.2.2). —** *Let $X$ be reduced, let $f : X \to Y$ be a morphism, and let $Z$ be a closed subprescheme of $Y$ such that $f(X) \subset Z$. Then $f$ factors as $X \xrightarrow{g} Z \xrightarrow{j} Y$, where $j$ is the injection morphism.*

*Proof.* It follows from the hypotheses that the closed subprescheme $f^{-1}(Z)$ of $X$ has all of $X$ as its underlying space (4.4.1); since $X$ is reduced, this closed subprescheme agrees with $X$ (5.1.2), and the proposition then follows from (4.4.1). □

[12] *[Trans.] The French source prints $\mathscr{F}|Y$ here; the restriction required on the neighbourhood $V$ is $\mathscr{F}|V$.*

**Corollary (5.2.3).** — *Let $X$ be a reduced subprescheme of a prescheme $Y$; if $Z$ is the closed reduced subprescheme of $Y$ that has $\overline{X}$ as its underlying space, then $X$ is a subprescheme induced on an open subset of $Z$.*

*Proof.* There is indeed an open subset $U$ of $Y$ such that $X = U \cap \overline{X}$; since, by (5.2.2), $X$ is a reduced subprescheme of $Z$, the subprescheme $X$ is induced by $Z$ on the open subspace $X$ by uniqueness (5.2.1). □ **I** | 132

**Corollary (5.2.4).** — *Let $f : X \to Y$ be a morphism, and $X'$ (resp. $Y'$) a closed subprescheme of $X$ (resp. $Y$) defined by a quasi-coherent sheaf of ideals $\mathscr{J}$ (resp. $\mathscr{K}$) of $\mathscr{O}_X$ (resp. $\mathscr{O}_Y$). Suppose that $X'$ is reduced, and that $f(X') \subset Y'$. Then $f^*(\mathscr{K})\mathscr{O}_X \subset \mathscr{J}$.*

*Proof.* Since, by (5.2.2), the restriction of $f$ to $X'$ factors as $X' \to Y' \to Y$, it suffices to apply (4.4.6). □

### 5.3. Diagonal; graph of a morphism

**(5.3.1).** Let $X$ be an $S$-prescheme; we define the *diagonal morphism* from $X$ into $X \times_S X$, denoted by $\Delta_{X|S}$, or $\Delta_X$, or even $\Delta$ if no confusion may arise, to be the $S$-morphism $(1_X, 1_X)_S$, or, in other words, the unique $S$-morphism $\Delta_X$ such that

(5.3.1.1)
$$p_1 \circ \Delta_X = p_2 \circ \Delta_X = 1_X,$$

where $p_1$ and $p_2$ are the projections of $X \times_S X$ (Definition (3.2.1)). If $f : T \to X$ and $g : T \to Y$ are $S$-morphisms, we immediately have that

(5.3.1.2)
$$(f, g)_S = (f \times_S g) \circ \Delta_{T|S}.$$

The reader will note that the preceding definition and the results stated in (5.3.1) to (5.3.8) are true in any category, *provided that the products used within exist in the category*.

**Proposition (5.3.2).** — *Let $X$ and $Y$ be $S$-preschemes; if we make the canonical identification between $(X \times Y) \times (X \times Y)$ and $(X \times X) \times (Y \times Y)$, then the morphism $\Delta_{X \times Y}$ is identified with $\Delta_X \times \Delta_Y$.*

*Proof.* Indeed, if $p_1 : X \times X \to X$ and $q_1 : Y \times Y \to Y$ are the projections onto the first component, then the projection onto the first component $(X \times Y) \times (X \times Y) \to X \times Y$ is identified with $p_1 \times q_1$, and we have
$$(p_1 \times q_1) \circ (\Delta_X \times \Delta_Y) = (p_1 \circ \Delta_X) \times (q_1 \circ \Delta_Y) = 1_{X \times Y}$$
and we can argue similarly for the projections onto the second component. □

**Corollary (5.3.4).** — *For every base extension $S' \to S$, $\Delta_{X_{(S')}}$ is canonically identified with $(\Delta_X)_{(S')}$.*

*Proof.* It suffices to remark that $(X \times_S X)_{(S')}$ is canonically identified with $X_{(S')} \times_{S'} X_{(S')}$ (3.3.10). □

**Proposition (5.3.5).** — *Let $X$ and $Y$ be $S$-preschemes, and $\phi : S \to T$ a morphism of preschemes, which lets us consider every $S$-prescheme as a $T$-prescheme. Let $f : X \to S$ and $g : Y \to S$[13] be the structure morphisms, $p$ and $q$ the projections of $X \times_S Y$, and $\pi = f \circ p = g \circ q$ the structure morphism $X \times_S Y \to S$. Then the diagram*

(5.3.5.1)
$$\begin{tikzcd} X \times_S Y \arrow[r, "(p,q)_T"] \arrow[d, "\pi"'] & X \times_T Y \arrow[d, "f \times_T g"] \\ S \arrow[r, "\Delta_{S|T}"] & S \times_T S \end{tikzcd}$$

*commutes, and identifies $X \times_S Y$ with the product of the $(S \times_T S)$-preschemes $S$ and $X \times_T Y$, and the projections with $\pi$ and $(p, q)_T$.* **I** | 133

[13][Trans.] The French source prints only $Y \to S$ here, omitting the name $g$; the following identity $\pi = f \circ p = g \circ q$ identifies this second structure morphism as $g$.

*Proof.* By (3.4.3), we are led to proving the corresponding proposition *in the category of sets*, replacing $X$, $Y$, and $S$ by $X(Z)_T$, $Y(Z)_T$, and $S(Z)_T$ (respectively), with $Z$ being an arbitrary $T$-prescheme. But, in the category of sets, the proof is immediate and left to the reader. □

**Corollary (5.3.6). —** *The morphism $(p,q)_T$ can be identified (letting $P = S \times_T S$) with $1_{X\times_T Y} \times_P \Delta_S$.*

*Proof.* This follows from (5.3.5) and (3.3.4). □

**Corollary (5.3.7). —** *If $f : X \to Y$ is an $S$-morphism, then the diagram*

$$\begin{tikzcd} X \arrow[r, "(1_X,f)_S"] \arrow[d, "f"'] & X \times_S Y \arrow[d, "f\times_S 1_Y"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

*commutes, and identifies $X$ with the product of the $(Y \times_S Y)$-preschemes $Y$ and $X \times_S Y$.*

*Proof.* It suffices to apply (5.3.5), replacing $S$ by $Y$, and $T$ by $S$, and noting that $X \times_Y Y = X$ (3.3.3). □

**Proposition (5.3.8). —** *For $f : X \to Y$ to be a monomorphism of preschemes, it is necessary and sufficient for $\Delta_{X|Y}$ to be an isomorphism from $X$ to $X \times_Y X$.*

*Proof.* Indeed, $f$ is a monomorphism if and only if, for every $Y$-prescheme $Z$, the corresponding map $f' : X(Z)_Y \to Y(Z)_Y$ is injective. Since $Y(Z)_Y$ consists of a single element, this is equivalent to saying that $X(Z)_Y$ has at most one element[14], or, equivalently, that the diagonal map from $X(Z)_Y$ to $X(Z)_Y \times X(Z)_Y$ is a bijection. The latter product is exactly the set $(X \times_Y X)(Z)_Y$ (3.4.3.1), so this is equivalent to $\Delta_{X|Y}$ being an isomorphism. □

**Proposition (5.3.9). —** *The diagonal morphism $\Delta_X$ is an immersion from $X$ to $X \times_S X$.*

*Proof.* Indeed, since the continuous maps $p_1$ and $\Delta_X$ from the underlying spaces are such that $p_1 \circ \Delta_X$ is the identity, $\Delta_X$ is a homeomorphism from $X$ to $\Delta_X(X)$. Similarly, the composite homomorphism $\mathscr{O}_x \to \mathscr{O}_{\Delta_X(x)} \to \mathscr{O}_x$ (composed of the homomorphisms corresponding to $p_1$ and $\Delta_X$) is the identity, which means that the homomorphism corresponding to $\Delta_X$ is surjective; the proposition thus follows from (4.2.2). □

We say that the subprescheme of $X \times_S X$ associated to the immersion $\Delta_X$ (4.2.1) is *the diagonal* of $X \times_S X$.

**Corollary (5.3.10). —** *Under the hypotheses of* (5.3.5), *$(p,q)_T$ is an immersion.*

*Proof.* This follows from (5.3.6) and (4.3.1). □

We say (under the hypotheses of (5.3.5)) that $(p,q)_T$ is the *canonical immersion* of $X \times_S Y$ into $X \times_T Y$.

**Corollary (5.3.11). —** *Let $X$ and $Y$ be $S$-preschemes, and $f : X \to Y$ an $S$-morphism; then the graph morphism $\Gamma_f = (1_X, f)_S$ of $f$* (3.3.14) *is an immersion of $X$ into $X \times_S Y$.*

*Proof.* This is a particular case of Corollary (5.3.10), where we replace $S$ by $Y$, and $T$ by $S$ (cf. (5.3.7)). □

The subprescheme of $X \times_S Y$ associated to the immersion $\Gamma_f$ (4.2.1) is called *the graph* of the morphism $f$; the subpreschemes of $X \times_S Y$ that are graphs of morphisms $X \to Y$ are characterised by the property that the restriction to such a subprescheme $G$ of the projection $p_1 : X \times_S Y \to X$ is an *isomorphism* $g$ from $G$ to $X$: $G$ is the graph of the morphism $p_2 \circ g^{-1}$, where $p_2$ is the projection $X \times_S Y \to Y$. I | 134

When we take, in particular, $X = S$, then the $S$-morphisms $S \to Y$ (which are exactly the $S$-*sections* of $Y$ (2.5.5)) are equal to their graph morphisms; the subpreschemes of $Y$ that are the graphs

[14] *[Trans.] The French says that $X(Z)_Y$ is likewise reduced to one element; injectivity into the singleton $Y(Z)_Y$ yields only at most one element, allowing the empty case.*

of $S$-sections (in other words, those that are isomorphic to $S$ by the restriction of the structure morphism $Y \to S$) are then also called the *images* of these sections, or, by an abuse of language, the *S-sections* of $Y$.

**Corollary (5.3.12). —** *With the hypotheses and notation of (5.3.11), for every morphism $g : S' \to S$, let $f'$ be the inverse image of $f$ under $g$ (3.3.7); then $\Gamma_{f'}$ is the inverse image of $\Gamma_f$ under $g$.*

*Proof.* This is a particular case of (3.3.10.1). ☐

**Corollary (5.3.13). —** *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms; if $g \circ f$ is an immersion (resp. a local immersion), then so too is $f$.*

*Proof.* Indeed, $f$ factors as $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$. Furthermore, $p_2$ can be identified with $(g \circ f) \times_Z 1_Y$ (3.3.4); if $g \circ f$ is an immersion (resp. a local immersion), then so too is $p_2$ ((4.3.1) and (4.5.5)), and since $\Gamma_f$ is an immersion (5.3.11), we are done, by (4.2.4) (resp. (4.5.5)). ☐

**Corollary (5.3.14). —** *Let $j : X \to Y$ and $g : X \to Z$ be $S$-morphisms. If $j$ is an immersion (resp. a local immersion), then so too is $(j, g)_S$.*

*Proof.* Indeed, if $p : Y \times_S Z \to Y$ is the projection onto the first component, then we have $j = p \circ (j, g)_S$, and it suffices to apply (5.3.13). ☐

**Proposition (5.3.15). —** *If $f : X \to Y$ is an $S$-morphism, then the diagram*

(5.3.15.1)
$$\begin{tikzcd} X \arrow[r, "\Delta_X"] \arrow[d, "f"'] & X \times_S X \arrow[d, "f \times_S f"] \\ Y \arrow[r, "\Delta_Y"] & Y \times_S Y \end{tikzcd}$$

*commutes* (in other words, $\Delta_X$ is a functorial morphism in the category of preschemes).

*Proof.* The proof is immediate and left to the reader. ☐

**Corollary (5.3.16). —** *If $X$ is a subprescheme of $Y$, then the diagonal $\Delta_X(X)$ can be identified with a subprescheme of $\Delta_Y(Y)$, and the underlying space can be identified with*

$$\Delta_Y(Y) \cap p_1^{-1}(X) = \Delta_Y(Y) \cap p_2^{-1}(X)$$

*($p_1$ and $p_2$ being the projections of $Y \times_S Y$).*

*Proof.* Applying (5.3.15) to the injection morphism $f : X \to Y$, we see that $f \times_S f$ is an immersion that identifies the underlying space of $X \times_S X$ with the subspace $p_1^{-1}(X) \cap p_2^{-1}(X)$ of $Y \times_S Y$ (4.3.1); further, if $z \in \Delta_Y(Y) \cap p_1^{-1}(X)$, then we have $z = \Delta_Y(y)$ and $y = p_1(z) \in X$, so $y = f(y)$, and $z = \Delta_Y(f(y))$ belongs to $\Delta_X(X)$ by the commutativity of (5.3.15.1). ☐ I | 135

**Corollary (5.3.17). —** *Let $f_1 : Y \to X$ and $f_2 : Y \to X$ be $S$-morphisms, and $y$ a point of $Y$ such that $f_1(y) = f_2(y) = x$, and such that the homomorphisms $k(x) \to k(y)$ corresponding to $f_1$ and $f_2$ are identical. Then, if $f = (f_1, f_2)_S$, the point $f(y)$ belongs to the diagonal $\Delta_{X|S}(X)$.*

*Proof.* The two homomorphisms $k(x) \to k(y)$ corresponding to $f_i$ ($i = 1, 2$) define two $S$-morphisms $g_i : \operatorname{Spec}(k(y)) \to \operatorname{Spec}(k(x))$ such that the diagrams

$$\begin{tikzcd} \operatorname{Spec}(k(y)) \arrow[r, "g_i"] \arrow[d] & \operatorname{Spec}(k(x)) \arrow[d] \\ Y \arrow[r, "f_i"] & X \end{tikzcd}$$

commute. The diagram

$$\begin{tikzcd} \operatorname{Spec}(k(y)) \arrow[r, "(g_1, g_2)_S"] \arrow[d] & \operatorname{Spec}(k(x)) \times_S \operatorname{Spec}(k(x)) \arrow[d] \\ Y \arrow[r, "(f_1, f_2)_S"] & X \times_S X \end{tikzcd}$$

thus also commutes. But it follows from the equality $g_1 = g_2$ that the image under $(g_1, g_2)_S$ of the unique point of $\mathrm{Spec}(k(y))$ belongs to the diagonal of $\mathrm{Spec}(k(x)) \times_S \mathrm{Spec}(k(x))$; the conclusion then follows from (5.3.15). □

### 5.4. Separated morphisms and separated preschemes

**Definition (5.4.1). —** We say that a morphism of preschemes $f : X \to Y$ is *separated* if the diagonal morphism $X \to X \times_Y X$ is a *closed* immersion; we then also say that $X$ is a *separated prescheme over* $Y$, or a $Y$*-scheme*. We say that a prescheme $X$ is separated if it is separated over $\mathrm{Spec}(\mathbf{Z})$; we then also say that $X$ is a *scheme*[15] *(cf.* (5.5.7)*)*.

By (5.3.9), for $X$ to be separated over $Y$, it is necessary and sufficient for $\Delta_X(X)$ to be a *closed subspace* of the underlying space of $X \times_Y X$.

**Proposition (5.4.2). —** *Let $S \to T$ be a separated morphism. If $X$ and $Y$ are $S$-preschemes, then the canonical immersion $X \times_S Y \to X \times_T Y$* (5.3.10) *is closed.*

*Proof.* Indeed, if we refer to the diagram in (5.3.5.1), we see that $(p, q)_T$ can be considered as being obtained from $\Delta_{S|T}$ by the base extension $f \times_T g : X \times_T Y \to S \times_T S$; the proposition then follows from (4.3.2). □

**Corollary (5.4.3). —** *Let $Y$ be an $S$-scheme, and $f : X \to Y$ an $S$-morphism. Then the graph morphism $\Gamma_f : X \to X \times_S Y$* (5.3.11) *is a closed immersion.*

*Proof.* This is a particular case of (5.4.2), where we replace $S$ by $Y$, and $T$ by $S$. □

**Corollary (5.4.4). —** *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms, with $g$ separated. If $g \circ f$ is a closed immersion, then so too is $f$.*

*Proof.* The proof using (5.4.3) is the same as that of (5.3.13) using (5.3.11). □

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**Corollary (5.4.5). —** *Let $Z$ be an $S$-scheme, and $j : X \to Y$ and $g : X \to Z$ $S$-morphisms. If $j$ is a closed immersion, then so too is $(j, g)_S : X \to Y \times_S Z$.*

*Proof.* The proof using (5.4.4) is the same as that of (5.3.14) using (5.3.13). □

**Corollary (5.4.6). —** *If $X$ is an $S$-scheme, then every $S$-section of $X$* (2.5.5) *is a closed immersion.*

*Proof.* If $\phi : X \to S$ is the structure morphism, and $\psi : S \to X$ an $S$-section of $X$, it suffices to apply (5.4.5) to $\phi \circ \psi = 1_S$. □

**Corollary (5.4.7). —** *Let $S$ be an integral prescheme with generic point $s$, and $X$ an $S$-scheme. If two $S$-sections $f$ and $g$ are such that $f(s) = g(s)$, then $f = g$.*

*Proof.* Indeed, if $x = f(s) = g(s)$, then the homomorphisms $k(x) \to k(s)$ corresponding to $f$ and $g$ are necessarily identical. If $h = (f, g)_S$, we thus deduce from (5.3.17) that $h(s)$ belongs to the diagonal $Z = \Delta_X(X)$; but since $S = \overline{\{s\}}$, and since $Z$ is closed by hypothesis, we have $h(S) \subset Z$. It then follows from (5.2.2) that $h$ factors as $S \to Z \to X \times_S X$, and we thus conclude that $f = g$, by definition of the diagonal. □

**Remark (5.4.8). —** If we suppose, conversely, that the conclusion of (5.4.3) is true when $f = 1_Y$, then we can conclude that $Y$ is separated over $S$; similarly, if we suppose that the conclusion of (5.4.5) applies to the two morphisms $Y \xrightarrow{\Delta_Y} Y \times_Z Y \xrightarrow{p_1} Y$, then we can conclude that $\Delta_Y$ is a closed immersion, and thus that $Y$ is separated over $Z$; finally, if we assume that the conclusion of (5.4.6) is true for the $Y$-section $\Delta_Y$ of the $Y$-prescheme $Y \times_S Y \to Y$, then this implies that $Y$ is separated over $S$.

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[15] *[Trans.] We repeat here the warning given at the very start of this translation: the early versions of the EGA use* prescheme *to mean what is now usually called a scheme, and* scheme *for what is now usually called a separated scheme.* Grothendieck himself later said that the more modern terminology was preferable, but we have decided to keep this translation 'historically accurate' by using the older nomenclature.

### 5.5. Separation criteria

**Proposition (5.5.1).** —

(i) *Every monomorphism of preschemes (and, in particular, every immersion) is a separated morphism.*
(ii) *The composition of any two separated morphisms is separated.*
(iii) *If $f : X \to X'$ and $g : Y \to Y'$ are separated $S$-morphisms, then $f \times_S g$ is separated.*
(iv) *If $f : X \to Y$ is a separated $S$-morphism, then the $S'$-morphism $f_{(S')}$ is separated for every extension $S' \to S$ of the base prescheme.*
(v) *If the composition $g \circ f$ is separated, then $f$ is separated.*
(vi) *For a morphism $f$ to be separated, it is necessary and sufficient for $f_{\text{red}}$ (5.1.5) to be separated.*

*Proof.* Note that (i) is an immediate consequence of (5.3.8). If $f : X \to Y$ and $g : Y \to Z$ are morphisms, then the diagram

(5.5.1.1)
$$\begin{tikzcd} X \arrow[rr, "\Delta_{X|Z}"] \arrow[dr, "\Delta_{X|Y}"'] & & X \times_Z X \\ & X \times_Y X \arrow[ur, "j"'] & \end{tikzcd}$$

commutes (where $j$ denotes the canonical immersion (5.3.10)), as can be immediately verified. If $f$ and $g$ are separated, then $\Delta_{X|Y}$ is a closed immersion by definition, and $j$ is a closed immersion by (5.4.2), whence $\Delta_{X|Z}$ is a closed immersion by (4.2.4), which proves (ii). Given (i) and (ii), (iii) **I** | 137
and (iv) are equivalent (3.5.1), so it suffices to prove (iv). But $X_{(S')} \times_{Y_{(S')}} X_{(S')}$ is canonically identified with $(X \times_Y X) \times_Y Y_{(S')}$ by (3.3.11) and (3.3.9.1), and we immediately see that the diagonal morphism $\Delta_{X_{(S')}}$ can then be identified with $\Delta_X \times_Y 1_{Y_{(S')}}$; the proposition then follows from (4.3.1).

To prove (v), consider, as in (5.3.13), the factorisation $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ of $f$, noting that $p_2 = (g \circ f) \times_Z 1_Y$; the hypothesis (that $g \circ f$ is separated) implies that $p_2$ is separated, by (iii) and (i), and, since $\Gamma_f$ is an immersion, $\Gamma_f$ is separated, by (i), whence $f$ is separated, by (ii). Finally, to prove (vi), recall that the preschemes $X_{\text{red}} \times_{Y_{\text{red}}} X_{\text{red}}$ and $X_{\text{red}} \times_Y X_{\text{red}}$ are canonically identified with one another (5.1.7); if we denote by $j$ the injection $X_{\text{red}} \to X$, then the diagram

$$\begin{tikzcd} X_{\text{red}} \arrow[r, "\Delta_{X_{\text{red}}}"] \arrow[d, "j"'] & X_{\text{red}} \times_Y X_{\text{red}} \arrow[d, "j \times_Y j"] \\ X \arrow[r, "\Delta_X"'] & X \times_Y X \end{tikzcd}$$

commutes (5.3.15), and the proposition follows from the fact that the vertical arrows are homeomorphisms of the underlying spaces (4.3.1). □

**Corollary (5.5.2).** — *If $f : X \to Y$ is separated, then the restriction of $f$ to any subprescheme of $X$ is separated.*

*Proof.* This follows from (5.5.1, i and ii). □

**Corollary (5.5.3).** — *If $X$ and $Y$ are $S$-preschemes such that $Y$ is separated over $S$, then $X \times_S Y$ is separated over $X$.*

*Proof.* This is a particular case of (5.5.1, iv). □

**Proposition (5.5.4).** — *Let $X$ be a prescheme, and assume that its underlying space is a* finite *union of closed subsets $X_k$ ($1 \leqslant k \leqslant n$); for each $k$, consider the reduced subprescheme of $X$ that has $X_k$ as its underlying space (5.2.1), and denote this again by $X_k$. Let $f : X \to Y$ be a morphism, and, for each $k$, let $Y_k$ be a closed subset of $Y$ such that $f(X_k) \subset Y_k$; we again denote by $Y_k$ the reduced subprescheme of $Y$ that has $Y_k$ as its underlying space, so that the restriction $X_k \to Y$ of $f$ to $X_k$ factors as $X_k \xrightarrow{f_k} Y_k \to Y$ (5.2.2). For $f$ to be separated, it is necessary and sufficient for all the $f_k$ to be separated.*

*Proof.* The necessity follows from (5.5.1, i, ii, and v). Conversely, if the condition of the statement is satisfied, then each of the restrictions $X_k \to Y$ of $f$ is separated (5.5.1, (i) and (ii)); if $p_1$ and $p_2$ are the projections of $X \times_Y X$, then the subspace $\Delta_{X_k}(X_k)$ can be identified with the subspace $\Delta_X(X) \cap p_1^{-1}(X_k)$ of the underlying space of $X \times_Y X$ (5.3.16); these subspaces are closed in $X \times_Y X$, and thus so too is their union $\Delta_X(X)$. □

Suppose, in particular, that the $X_k$ are the *irreducible components* of $X$; then we can suppose that the $Y_k$ are the irreducible components of $Y$ (**0**, 2.1.5); Proposition (5.5.4) therefore reduces, in this case, the notion of separation to the case of *integral* preschemes (2.1.7).

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**Proposition (5.5.5).** — *Let $(Y_\lambda)$ be an open cover of a prescheme $Y$; for a morphism $f : X \to Y$ to be separated, it is necessary and sufficient for each of its restrictions $f^{-1}(Y_\lambda) \to Y_\lambda$ to be separated.*

*Proof.* If we set $X_\lambda = f^{-1}(Y_\lambda)$, everything reduces, by taking (4.2.4, b) and the identification of the products $X_\lambda \times_Y X_\lambda$ and $X_\lambda \times_{Y_\lambda} X_\lambda$ (3.2.5) into account, to proving that the $X_\lambda \times_Y X_\lambda$ form a cover of $X \times_Y X$. But if we set $Y_{\lambda\mu} = Y_\lambda \cap Y_\mu$ and $X_{\lambda\mu} = X_\lambda \cap X_\mu = f^{-1}(Y_{\lambda\mu})$, then $X_\lambda \times_Y X_\mu$ can be identified with the product $X_{\lambda\mu} \times_{Y_{\lambda\mu}} X_{\lambda\mu}$ (3.2.6.4), and so also with $X_{\lambda\mu} \times_Y X_{\lambda\mu}$ (3.2.5), and finally with an open subset of $X_\lambda \times_Y X_\lambda$, which proves our claim (3.2.7). □

Proposition (5.5.4) allows us, by taking an affine open cover of $Y$, to reduce the study of separated morphisms to that of separated morphisms taking values in affine schemes.

**Proposition (5.5.6).** — *Let $Y$ be an* affine *scheme, $X$ a prescheme, and $(U_\alpha)$ a cover of $X$ by affine open subsets. For a morphism $f : X \to Y$ to be separated, it is necessary and sufficient for $U_\alpha \cap U_\beta$ to be, for every pair of indices $(\alpha, \beta)$, an affine open subset,* and *for the ring $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X)$ to be generated by the union of the canonical images of the rings $\Gamma(U_\alpha, \mathscr{O}_X)$ and $\Gamma(U_\beta, \mathscr{O}_X)$.*

*Proof.* The $U_\alpha \times_Y U_\beta$ form an open cover of $X \times_Y X$ (3.2.7); denoting the projections of $X \times_Y X$ by $p$ and $q$, we have

$$\begin{aligned}\Delta_X^{-1}(U_\alpha \times_Y U_\beta) &= \Delta_X^{-1}(p^{-1}(U_\alpha) \cap q^{-1}(U_\beta)) \\ &= \Delta_X^{-1}(p^{-1}(U_\alpha)) \cap \Delta_X^{-1}(q^{-1}(U_\beta)) = U_\alpha \cap U_\beta;\end{aligned}$$

so everything reduces to proving that the restriction of $\Delta_X$ to $U_\alpha \cap U_\beta$ is a closed immersion into $U_\alpha \times_Y U_\beta$. But this restriction is exactly $(j_\alpha, j_\beta)_Y$, where $j_\alpha$ (resp. $j_\beta$) denotes the injection morphism from $U_\alpha \cap U_\beta$ to $U_\alpha$ (resp. $U_\beta$), as follows from the definitions. Since $U_\alpha \times_Y U_\beta$ is an affine scheme whose ring is canonically isomorphic to $\Gamma(U_\alpha, \mathscr{O}_X) \otimes_{\Gamma(Y, \mathscr{O}_Y)} \Gamma(U_\beta, \mathscr{O}_X)$ (3.2.2), we see that $U_\alpha \cap U_\beta$ must be an affine scheme, and that the map $h_\alpha \otimes h_\beta \mapsto h_\alpha h_\beta$ from the ring $A(U_\alpha \times_Y U_\beta)$ to $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X)$ must be surjective (4.2.3), which finishes the proof. □

**Corollary (5.5.7).** — *An affine scheme is separated* (and is thus a *scheme*, which justifies the terminology of (5.4.1)).

**Corollary (5.5.8).** — *Let $Y$ be an affine scheme; for $f : X \to Y$ to be a separated morphism, it is necessary and sufficient for $X$ to be separated* (in other words, for $X$ to be a *scheme*).

*Proof.* Indeed, we see that the criterion of (5.5.6) does not depend on $f$. □

**Corollary (5.5.9).** — *For a morphism $f : X \to Y$ to be separated, it is necessary for the induced prescheme $f^{-1}(U)$ to be separated for every open subset $U$ on which $Y$ induces a separated prescheme, and it is sufficient for this to hold for every affine open subset $U \subset Y$.*

*Proof.* The necessity of the condition follows from (5.5.4) and (5.5.1, ii); the sufficiency follows from (5.5.4) and (5.5.8), taking into account the existence of affine open covers of $Y$. □

In particular, if $X$ and $Y$ are affine schemes, then *every* morphism $X \to Y$ is separated.

**Proposition (5.5.10).** — *Let $Y$ be a* scheme, *and $f : X \to Y$ a morphism. For every affine open subset $U$ of $X$, and every affine open subset $V$ of $Y$, $U \cap f^{-1}(V)$ is affine.*

*Proof.* Let $p_1$ and $p_2$ be the projections of $X \times_{\mathbf{Z}} Y$; the subspace $U \cap f^{-1}(V)$ is the image of $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ under $p_1$. But $p_1^{-1}(U) \cap p_2^{-1}(V)$ can be identified with the underlying space of the prescheme $U \times_{\mathbf{Z}} V$ (3.2.7), and is thus an affine scheme (3.2.2); since $\Gamma_f(X)$ is closed in **I** | 139
$X \times_{\mathbf{Z}} Y$ (5.4.3), $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ is closed in $U \times_{\mathbf{Z}} V$, and so the prescheme induced by the subprescheme of $X \times_{\mathbf{Z}} Y$ associated to $\Gamma_f$ (4.2.1) on the open subset $\Gamma_f(X) \cap p_1^{-1}(U) \cap p_2^{-1}(V)$ of its underlying space is a closed subprescheme of an affine scheme, and thus an affine scheme (4.2.3). The proposition then follows from the fact that $\Gamma_f$ is an immersion. ☐

**Examples (5.5.11).** — The prescheme from Example (2.3.2) ("the projective line over a field $K$") is *separated*, because, for the cover $(X_1, X_2)$ of $X$ by affine open subsets, $X_1 \cap X_2 = U_{12}$ is affine, and $\Gamma(U_{12}, \mathscr{O}_X)$, the ring of rational fractions of the form $f(s)/s^m$ with $f \in K[s]$, is generated by $K[s]$ and $1/s$, so the conditions of (5.5.6) are satisfied.

With the same choice of $X_1$, $X_2$, $U_{12}$, and $U_{21}$ as in Example (2.3.2), now take $u_{12}$ to be the isomorphism which sends $f(s)$ to $f(t)$; we now obtain, by gluing, a *non-separated integral* prescheme $X$, because the first condition of (5.5.6) is satisfied, but not the second. It is immediate here that $\Gamma(X, \mathscr{O}_X) \to \Gamma(X_1, \mathscr{O}_X) = K[s]$ is an isomorphism; the inverse isomorphism defines a morphism $f : X \to \operatorname{Spec}(K[s])$ that is surjective, and for every $y \in \operatorname{Spec}(K[s])$ such that $\mathfrak{j}_y \neq (s)$, $f^{-1}(y)$ consists of a single point, but for $\mathfrak{j}_y = (s)$, $f^{-1}(y)$ consists of *two* distinct points[16] (we say that $X$ is the "affine line over $K$ with the point 0 doubled").

We can also give examples where *neither* of the two conditions of (5.5.6) is satisfied. First note that, in the prime spectrum $Y$ of the ring $A = K[s,t]$ of polynomials in two indeterminates over a field $K$, the open subset $U$ given by the union of $D(s)$ and $D(t)$ is *not an affine open subset*. Indeed, if $z$ is a section of $\mathscr{O}_Y$ over $U$, there exist two integers $m, n \geqslant 0$ such that $s^m z$ and $t^n z$ are the restrictions of polynomials in $s$ and $t$ to $U$ (1.4.1), which is clearly possible only if the section $z$ extends to a section over the whole of $Y$, identified with a polynomial in $s$ and $t$. If $U$ were an affine open subset, then the injection morphism $U \to Y$ would be an isomorphism (1.7.3), which is a contradiction, since $U \neq Y$.

With the above in mind, take two affine schemes $Y_1$ and $Y_2$, prime spectra of the rings $A_1 = K[s_1, t_1]$ and $A_2 = K[s_2, t_2]$ (respectively); take $U_{12} = D(s_1) \cup D(t_1)$ and $U_{21} = D(s_2) \cup D(t_2)$, and take $u_{12}$ to be the restriction of an isomorphism $Y_2 \to Y_1$ to $U_{21}$ corresponding to the isomorphism of rings that sends $f(s_1, t_1)$ to $f(s_2, t_2)$; we then have an example where neither condition of (5.5.6) is satisfied (the integral prescheme thus obtained is called "the affine plane over $K$ with the point 0 doubled").

**Remark (5.5.12).** — Given some property **P** of morphisms of preschemes, consider the following propositions.

(i) *Every closed immersion has property* ***P***.
(ii) *The composition of any two morphisms that both have property* ***P*** *also has property* ***P***.
(iii) *If $f : X \to X'$ and $g : Y \to Y'$ are $S$-morphisms that have property* ***P***, *then $f \times_S g$ has property* ***P***. **I** | 140
(iv) *If $f : X \to Y$ is an $S$-morphism that has property* ***P***, *then every $S'$-morphism $f_{(S')}$ obtained by an extension $S' \to S$ of the base prescheme also has property* ***P***.
(v) *If the composition $g \circ f$ of two morphisms $f : X \to Y$ and $g : Y \to Z$ has property* ***P***, *and $g$ is separated, then $f$ has property* ***P***.
(vi) *If a morphism $f : X \to Y$ has property* ***P***, *then so too does $f_{\mathrm{red}}$* (5.1.5).

If we suppose that (i) and (ii) are both true, then (iii) and (iv) are *equivalent*, and (v) and (vi) are *consequences* of (i), (ii), and (iii).

The first claim has already been shown (3.5.1). Consider the factorisation (5.3.13) $X \xrightarrow{\Gamma_f} X \times_Z Y \xrightarrow{p_2} Y$ of $f$; the equation $p_2 = (g \circ f) \times_Z 1_Y$ shows that, if $g \circ f$ has property **P**, then so too does $p_2$, by (iii); if $g$ is separated, then $\Gamma_f$ is a closed immersion (5.4.3), and so also has property **P**, by (i); finally, by (ii), $f$ has property **P**.

[16] *[Trans.] The French source prints* $(0)$ *in both ideal conditions. For the doubled origin in* $\operatorname{Spec}(K[s])$, *the relevant prime ideal is* $(s)$; $(0)$ *is the generic point.*

Finally, consider the commutative diagram

$$\begin{tikzcd} X_{\mathrm{red}} \arrow[r, "f_{\mathrm{red}}"] \arrow[d] & Y_{\mathrm{red}} \arrow[d] \\ X \arrow[r, "f"] & Y, \end{tikzcd}$$

where the vertical arrows are the closed immersions (5.1.5), and thus have property **P**, by (i). The hypothesis that $f$ has property **P** implies, by (ii), that $X_{\mathrm{red}} \xrightarrow{f_{\mathrm{red}}} Y_{\mathrm{red}} \to Y$ has property **P**; finally, since a closed immersion is separated (5.5.1, i), $f_{\mathrm{red}}$ has property **P**, by (v).

Note that, if we consider the propositions

(i′) *Every immersion has property **P**;*

(v′) *If $g \circ f$ has property **P**, then so too does $f$;*

then the above arguments show that (v′) is a consequence of (i′), (ii), and (iii).

**(5.5.13).** Note that (v) and (vi) are again consequences of (i), (iii), and

(ii′) *If $j : X \to Y$ is a closed immersion, and $g : Y \to Z$ is a morphism that has property **P**, then $g \circ j$ has property **P**.*

Similarly, (v′) is a consequence of (i′), (iii), and

(ii″) *If $j : X \to Y$ is an immersion, and $g : Y \to Z$ is a morphism that has property **P**, then $g \circ j$ has property **P**.*

This follows immediately from the arguments of (5.5.12).

## §6. Finiteness conditions

### 6.1. Noetherian and locally Noetherian preschemes

**Definition (6.1.1). —** We say that a prescheme $X$ is Noetherian (resp. locally Noetherian) if it is a finite union (resp. union) of affine open subsets $V_\alpha$ such that the ring of the scheme induced on each $V_\alpha$ is Noetherian.

It follows immediately from (1.5.2) that, if $X$ is locally Noetherian, then the structure sheaf $\mathscr{O}_X$ is a *coherent sheaf of rings*, since the question is a local one. Every *quasi-coherent $\mathscr{O}_X$-submodule* (resp. quasi-coherent quotient $\mathscr{O}_X$-module) of a *coherent* $\mathscr{O}_X$-module $\mathscr{F}$ is *coherent*, as the question is once again a local one, and it suffices to apply (1.5.1), (1.4.1), and (1.3.10), combined with the fact that a submodule (resp. quotient module) of a module of finite type over a Noetherian ring is of finite type. In particular, every *quasi-coherent sheaf of ideals* of $\mathscr{O}_X$ is *coherent*. **I** | 141

If a prescheme $X$ is a finite union (resp. union) of open subsets $W_\lambda$ in such a way that the preschemes induced on the $W_\lambda$ are Noetherian (resp. locally Noetherian), it is clear that $X$ is Noetherian (resp. locally Noetherian).

**Proposition (6.1.2). —** *For a prescheme $X$ to be Noetherian, it is necessary and sufficient for it to be locally Noetherian and have a quasi-compact underlying space. The underlying space itself is then also Noetherian.*

*Proof.* The first claim follows immediately from the definitions and (1.1.10, ii). The second follows from (1.1.6) and the fact that every space that is a finite union of Noetherian subspaces is itself Noetherian (**0**, 2.2.3). □

**Proposition (6.1.3). —** *Let $X$ be an affine scheme given by a ring $A$. The following conditions are equivalent:* (a) *$X$ is Noetherian;* (b) *$X$ is locally Noetherian;* (c) *$A$ is Noetherian.*

*Proof.* The equivalence between (a) and (b) follows from (6.1.2) and the fact that the underlying space of every affine scheme is quasi-compact (1.1.10); it is furthermore clear that (c) implies (a). To see that (a) implies (c), we remark that there is a finite cover $(V_i)$ of $X$ by affine open subsets such that the ring $A_i$ of the prescheme induced on $V_i$ is Noetherian. So let $(\mathfrak{a}_n)$ be an increasing sequence of ideals of $A$; under the canonical bijective correspondence (1.3.7), there corresponds to it an increasing sequence $(\widetilde{\mathfrak{a}}_n)$ of sheaves of ideals in $\widetilde{A} \equiv \mathscr{O}_X$; to see that the sequence $(\mathfrak{a}_n)$ is stationary, it suffices to prove that the sequence $(\widetilde{\mathfrak{a}}_n)$ is. But the restriction $\widetilde{\mathfrak{a}}_n|V_i$ is a quasi-coherent sheaf of ideals of $\mathscr{O}_X|V_i$, being the inverse image of $\widetilde{\mathfrak{a}}_n$ under the canonical injection $V_i \to X$ (**0**, 5.1.4); $\widetilde{\mathfrak{a}}_n|V_i$ is thus of the form $\widetilde{\mathfrak{a}}_{ni}$, where $\mathfrak{a}_{ni}$ is an ideal of $A_i$ (1.3.7). Since $A_i$ is Noetherian, the sequence $(\mathfrak{a}_{ni})$ is stationary for all $i$, whence the proposition. □

We note that the above argument proves also that *if $X$ is a Noetherian prescheme, then every increasing sequence of coherent sheaves of ideals of $\mathscr{O}_X$ is stationary.*

**Proposition (6.1.4).** — *Every subprescheme of a Noetherian (resp. locally Noetherian) prescheme is Noetherian (resp. locally Noetherian).*

*Proof.* It suffices to give a proof for a Noetherian prescheme $X$; further, by definition (6.1.1), we can also restrict to the case where $X$ is an affine scheme. Since every subprescheme of $X$ is a closed subprescheme of a prescheme induced on an open subset (4.1.3), we can restrict to the case of a subprescheme $Y$, either closed or induced on an open subset of $X$. The proof in the case where $Y$ is closed is immediate, since if $A$ is the ring of $X$, we know that $Y$ is an affine scheme given by the ring $A/\mathfrak{J}$, where $\mathfrak{J}$ is an ideal of $A$ (4.2.3); since $A$ is Noetherian (6.1.3), so too is $A/\mathfrak{J}$.

Now suppose that $Y$ is open in $X$; the underlying space of $Y$ is Noetherian (6.1.2), hence quasi-compact, and thus a finite union of open subsets $D(f_i)$ ($f_i \in A$); everything reduces to showing the proposition in the case where $Y = D(f)$ with $f \in A$. But then $Y$ is an affine scheme whose ring is isomorphic to $A_f$ (1.3.6); since $A$ is Noetherian (6.1.3), so too is $A_f$. □ **I** | 142

**(6.1.5).** We note that the *product* of two Noetherian $S$-preschemes is not necessarily Noetherian, even if the preschemes are affine, since the tensor product of two Noetherian algebras is not necessarily a Noetherian ring (cf. (6.3.8)).

**Proposition (6.1.6).** — *If $X$ is a Noetherian prescheme, the nilradical $\mathscr{N}_X$ of $\mathscr{O}_X$ is nilpotent.*

*Proof.* We can in fact cover $X$ with a finite number of affine open subsets $U_i$, and it suffices to prove that there exist integers $n_i$ such that $(\mathscr{N}_X|U_i)^{n_i} = 0$; if $n$ is the largest of the $n_i$, then we will have $\mathscr{N}_X^n = 0$. We can thus restrict to the case where $X = \operatorname{Spec}(A)$ is affine, with $A$ a Noetherian ring; by (5.1.1) and (1.3.13), it suffices to observe that the nilradical of $A$ is nilpotent ([11, p. 127, cor. 4]). □

**Corollary (6.1.7).** — *Let $X$ be a Noetherian prescheme; for $X$ to be an affine scheme, it is necessary and sufficient that $X_{\text{red}}$ be affine.*

*Proof.* This follows from (6.1.6) and (5.1.10). □

**Lemma (6.1.8).** — *Let $X$ be a topological space, $x$ a point of $X$, and $U$ an open neighbourhood of $x$ having only a finite number of irreducible components. Then there exists a neighbourhood $V$ of $x$ such that every open neighbourhood of $x$ contained in $V$ is connected.*

*Proof.* Let $U_i$ ($1 \leqslant i \leqslant m$) be the irreducible components of $U$ not containing $x$; the complement (in $U$) of the union of the $U_i$ is an open neighbourhood $V$ of $x$ inside $U$, and thus so too in $X$; it is also, incidentally, the complement (in $X$) of the union of the irreducible components of $X$ that do not contain $x$ (**0**, 2.1.6). So let $W$ be an open neighbourhood of $x$ contained in $V$. The irreducible components of $W$ are the intersections with $W$ of the irreducible components of $U$ that meet $W$ (**0**, 2.1.6), so these components contain $x$; since they are connected, so too is $W$. □

**Corollary (6.1.9).** — *A locally Noetherian topological space is locally connected (which implies, amongst other things, that its connected components are open).*

**Proposition (6.1.10).** — *Let $X$ be a locally Noetherian topological space. The following conditions are equivalent.*

(a) *The irreducible components of $X$ are open.*
(b) *The irreducible components of $X$ are exactly its connected components.*
(c) *The connected components of $X$ are irreducible.*
(d) *Two distinct irreducible components of $X$ have an empty intersection.*

*Finally, if $X$ is a prescheme, then these conditions are also equivalent to*

(e) *For every $x \in X$, $\operatorname{Spec}(\mathscr{O}_x)$ is irreducible (or, in other words, the nilradical of $\mathscr{O}_x$ is prime).*

*Proof.* It is immediate that (a) implies (b), because an irreducible space is connected, and (a) implies that the irreducible components of $X$ are the sets that are both open and closed. It is trivial that (b) implies (c); conversely, a closed set $F$ containing a connected component $C$ of $X$, with $C$ distinct **I** | 143
from $F$, cannot be irreducible, because not being connected means that $F$ is the union of two disjoint nonempty sets that are both open and closed in $F$, and thus closed in $X$; as a result, (c) implies (b). We immediately conclude from this that (c) implies (d), since two distinct connected components have no points in common.

We have not yet used the fact that $X$ is locally Noetherian. Suppose now that this is indeed the case, and we will show that (d) implies (a): by (**0**, 2.1.6), we can restrict ourselves to the case where the space $X$ is Noetherian, and so has only a finite number of irreducible components. Since they are closed and pairwise disjoint, they are open.

Finally, the equivalence between (d) and (e) holds true even without the assumption that the underlying space of the prescheme $X$ is locally Noetherian. We can in fact restrict ourselves to the case where $X = \operatorname{Spec}(A)$ is affine, by (**0**, 2.1.6); to say that $x$ is contained in only one single irreducible component of $X$ is to say that $\mathfrak{j}_x$ contains only one single minimal ideal of $A$ (1.1.14), which is equivalent to saying that $\mathfrak{j}_x \mathscr{O}_x$ contains only one single minimal ideal of $\mathscr{O}_x$, whence the conclusion. □

**Corollary (6.1.11).** — *Let $X$ be a locally Noetherian space. For $X$ to be irreducible, it is necessary and sufficient that $X$ be connected and nonempty, and that any two distinct irreducible components of $X$ have an empty intersection. If $X$ is a prescheme, this latter condition is equivalent to asking that* $\operatorname{Spec}(\mathscr{O}_x)$ *be irreducible for all $x \in X$.*

*Proof.* The second claim has already been shown in (6.1.10); the only thing thus remaining to show is that the conditions in the first claim are sufficient. But by (6.1.10), these conditions imply that the irreducible components of $X$ are exactly its connected components, and since $X$ is connected and nonempty, it is irreducible. □

**Corollary (6.1.12).** — *Let $X$ be a locally Noetherian prescheme. For $X$ to be integral, it is necessary and sufficient that $X$ be connected and that $\mathscr{O}_x$ be integral for all $x \in X$.*

**Proposition (6.1.13).** — *Let $X$ be a locally Noetherian prescheme, and let $x \in X$ be a point such that the nilradical $\mathscr{N}_x$ of $\mathscr{O}_x$ is prime (resp. such that $\mathscr{O}_x$ is reduced, resp. integral); then there exists an open neighbourhood $U$ of $x$ that is irreducible (resp. reduced, resp. integral).*

*Proof.* It suffices to consider two cases: where $\mathscr{N}_x$ is prime, and where $\mathscr{N}_x = 0$; the third hypothesis is a combination of the first two. If $\mathscr{N}_x$ is prime, then $x$ belongs to only one single irreducible component $Y$ of $X$ (6.1.10); the union of the irreducible components of $X$ that do not contain $x$ is closed (the set of these components being locally finite), and the complement $U$ of this union is thus open and contained in $Y$, and thus irreducible (**0**, 2.1.6). If $\mathscr{N}_x = 0$, we also have $\mathscr{N}_y = 0$ for any $y$ in a neighbourhood of $x$, because $\mathscr{N}$ is quasi-coherent (5.1.1), and thus coherent, since $X$ is locally Noetherian, and the conclusion then follows from (**0**, 5.2.2). □

### 6.2. Artinian preschemes

**Definition (6.2.1).** — We say that a prescheme is *Artinian* if it is affine and its ring is Artinian.